

The deep roots of human biology I

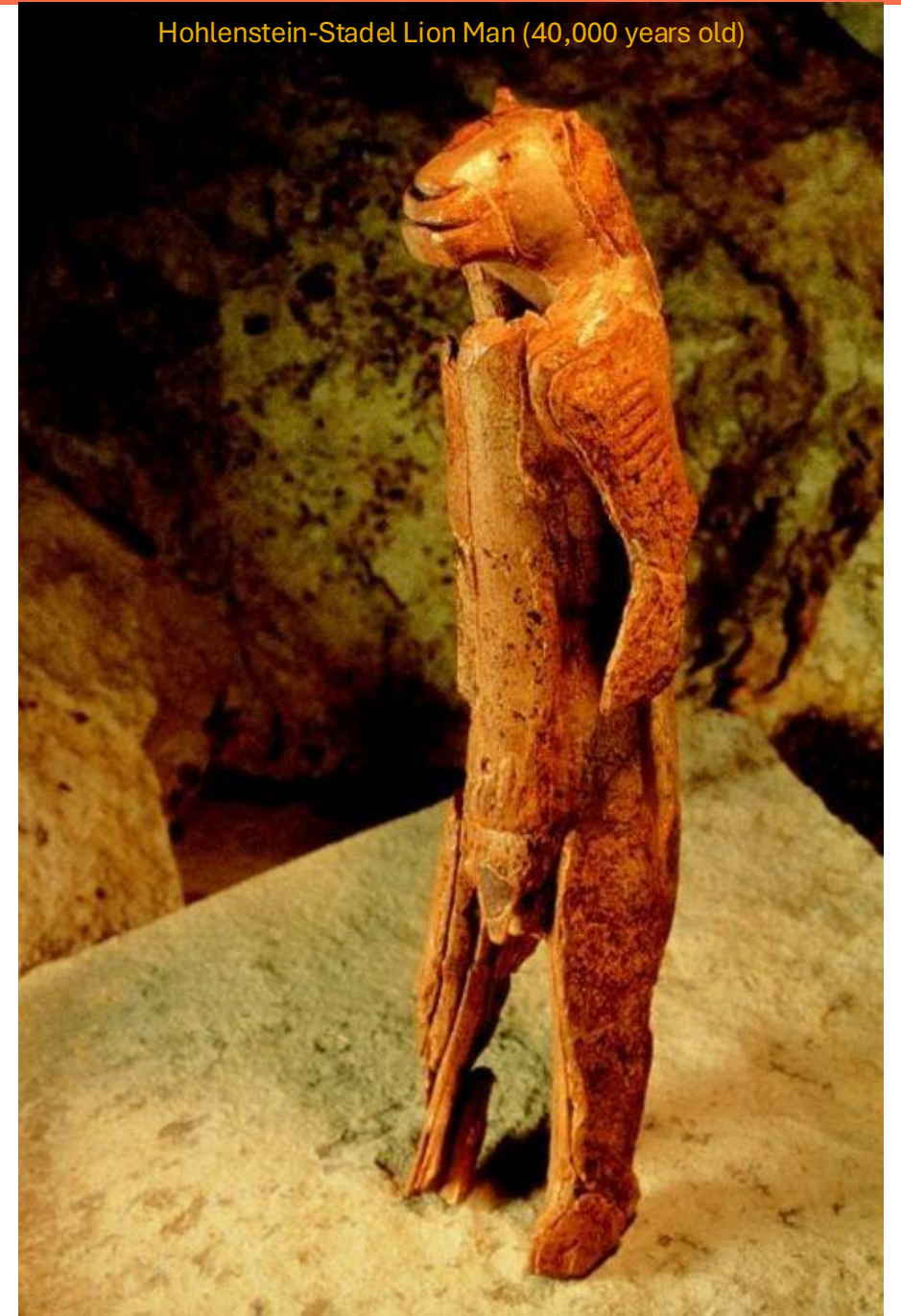
Dr Laura van Holstein

lavanho@emory.edu

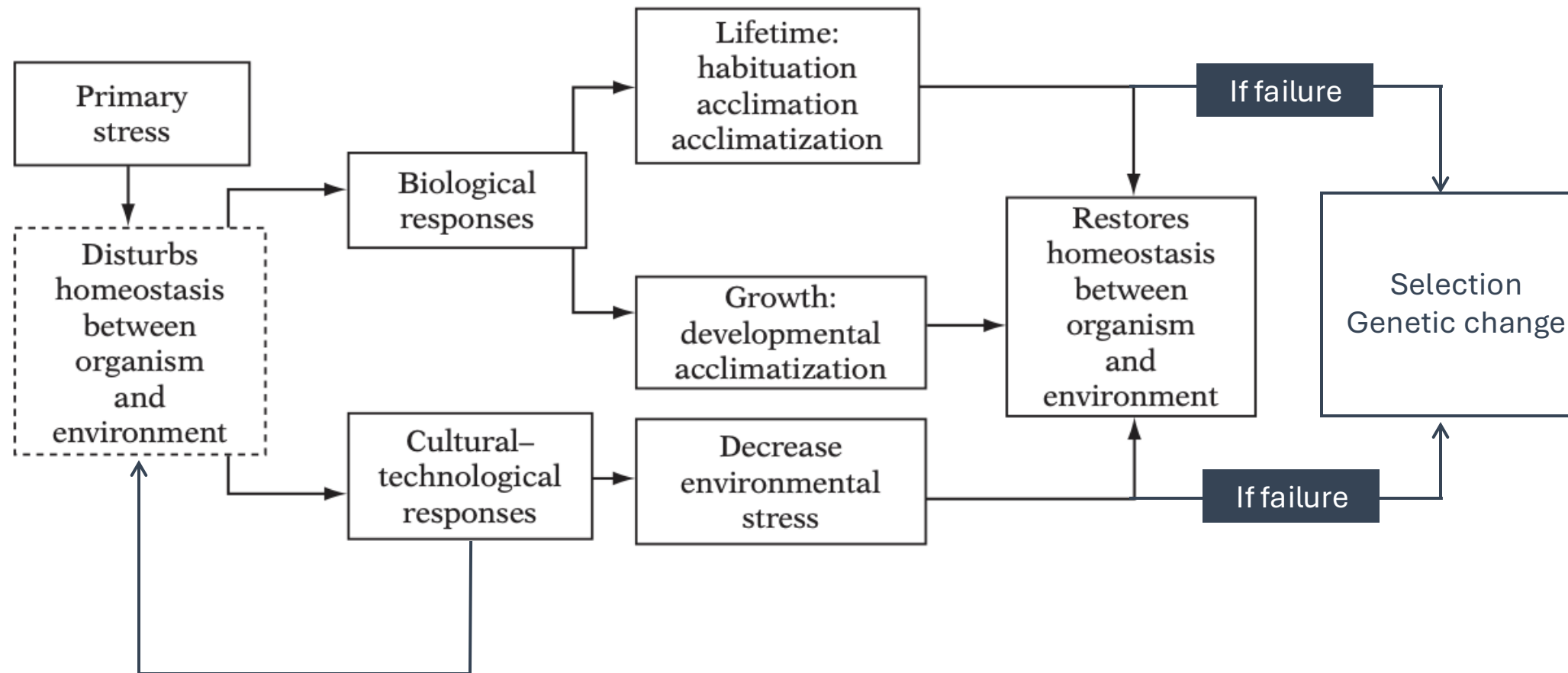
Anthropology 215

Office hours: Wednesdays 9-10.30AM

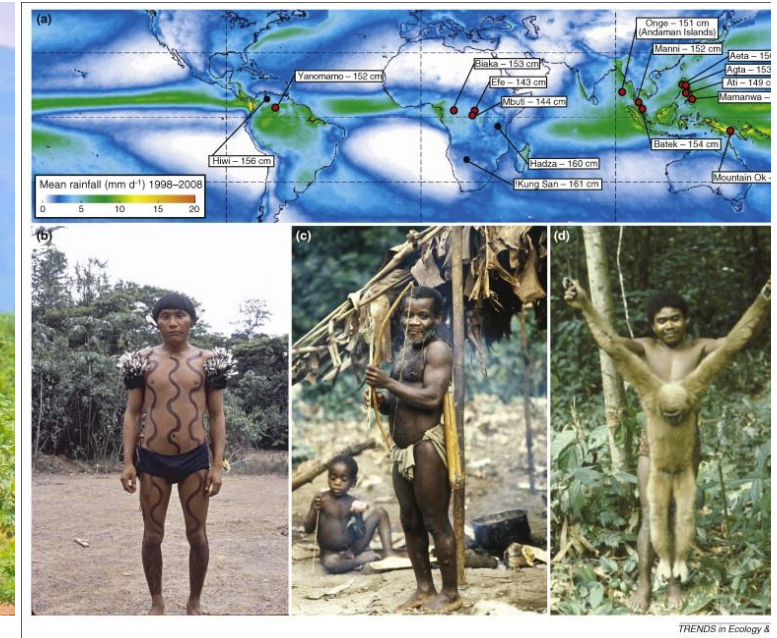
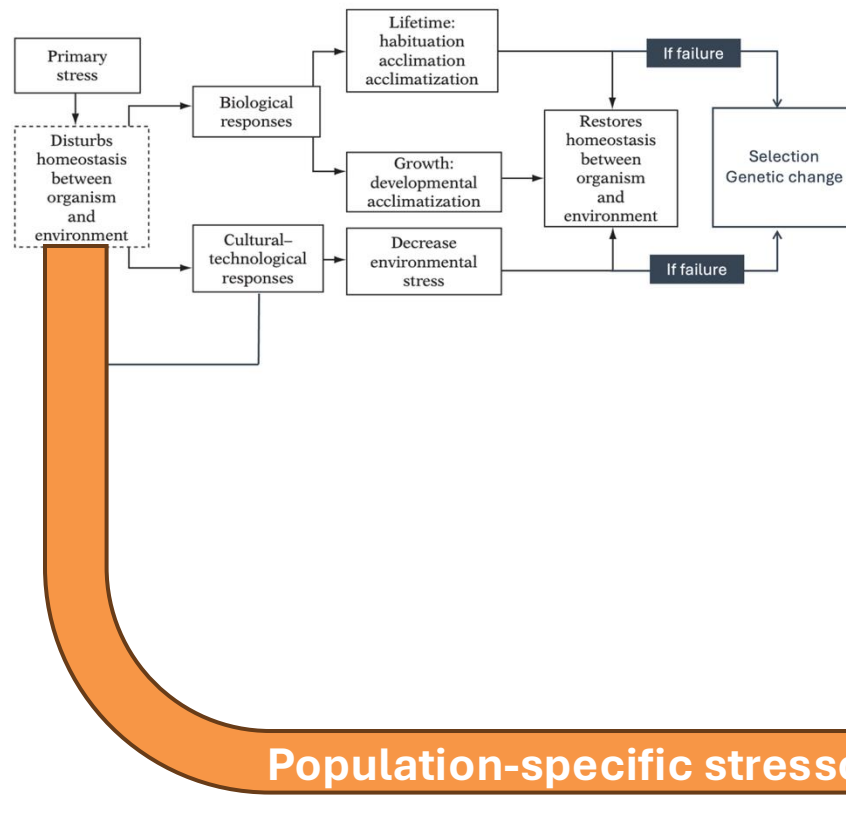
Hohlenstein-Stadel Lion Man (40,000 years old)



Picking up where we left off...



Picking up where we left off...

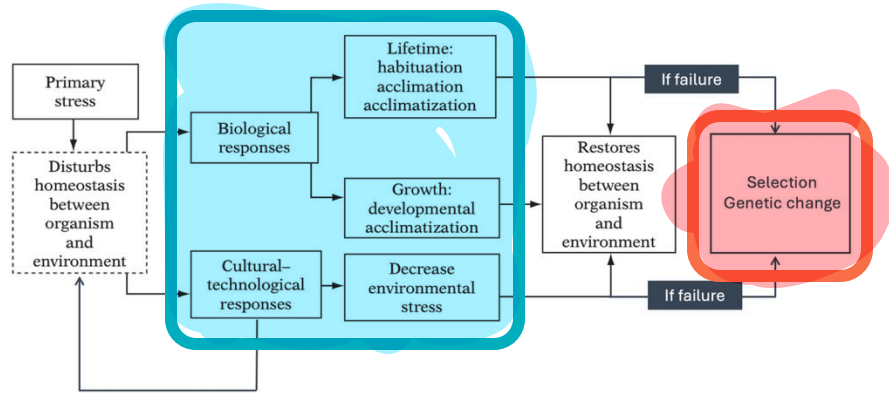


Adaptive phenotypic differences between populations

- Can be genetic, or not!
- Identifying causal process is major area of research in HEB

...but what about the traits that are **invariant**?

Traits that 'make' us human



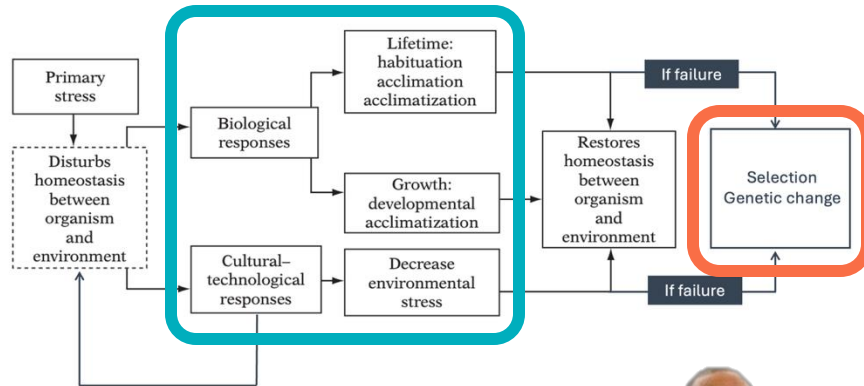
Invariant traits must either

- be **plastic** (non-genetic) and be driven by exactly the **same stressor** experienced across the world, or
- reflect **ancient** genetic adaptation to **long-term stressors** – that is, **selection**

Traits that all humans share
Don't vary across humans

Monotony?

Traits that 'make' us human

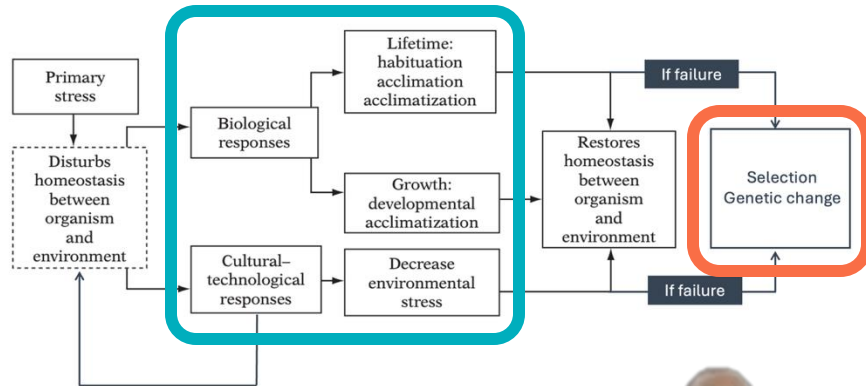


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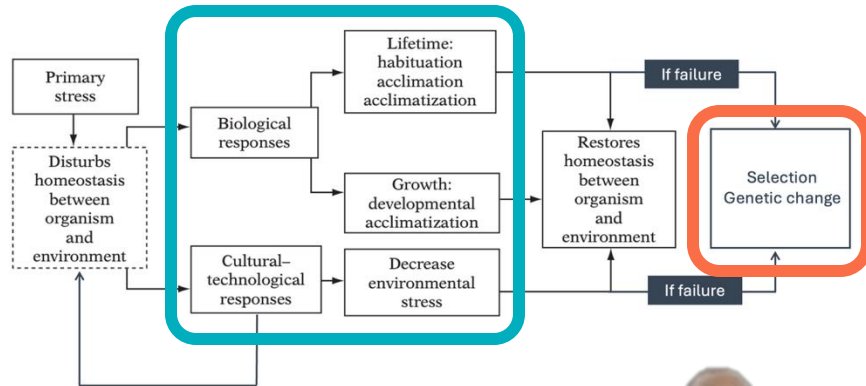
INVARIANT TRAITS THAT DEFINE HUMAN BIOLOGY

DNA	Metabolism	Sexual reproduction	Mitochondria	Central nervous system	Pentadactyly	Vertebrate jaw	Nails
RNA	Respiration		Muscle and nerve cells	Eyes		Mammalian ear	Stereoscopic vision
Cells	Glycolysis		Endoskeleton			Endothermy	Altriciality
ATP	Krebs cycle					Live birth	2.1.2.3
	Oxidative phosphorylation					Heterodonty	

Traits we share w/ all apes (not just great apes)

We share these with other mammals

Traits that 'make' us human



Invariant traits must either

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	Oxidative phosphorylation					Heterodonty	

To understand human biology, take a paleontology (or vet med) class!

Traits that 'make' us human



10 Ma

7 Ma

split between humans
and apes

- Obligate terrestrial bipedalism - *we are usually found on ground 2 legs*
Down sloped to this - how we get around
 - Bowl-shaped pelvis
 - Elongated femur
 - Flexible lower back
 - Highly derived feet
 - Inferiorly placed foramen magnum
- Gracility - *weak + small bones compared to other great apes*
- High activity levels - *less energetically not so much as other great apes*
- Chins
- Reduced canines; no Cp3 honing complex
- Precision grip
 - Straight phalanges (fingers)
- High encephalization quotient *large brains for body size*
- Globular cranium
- High capacity for plasticity (?) *questionable assertion*
- Long life histories with extended childhood & extended postreproductive period in females

Traits that 'make' us human



- Obligate terrestrial bipedalism
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Traits that 'make' us human



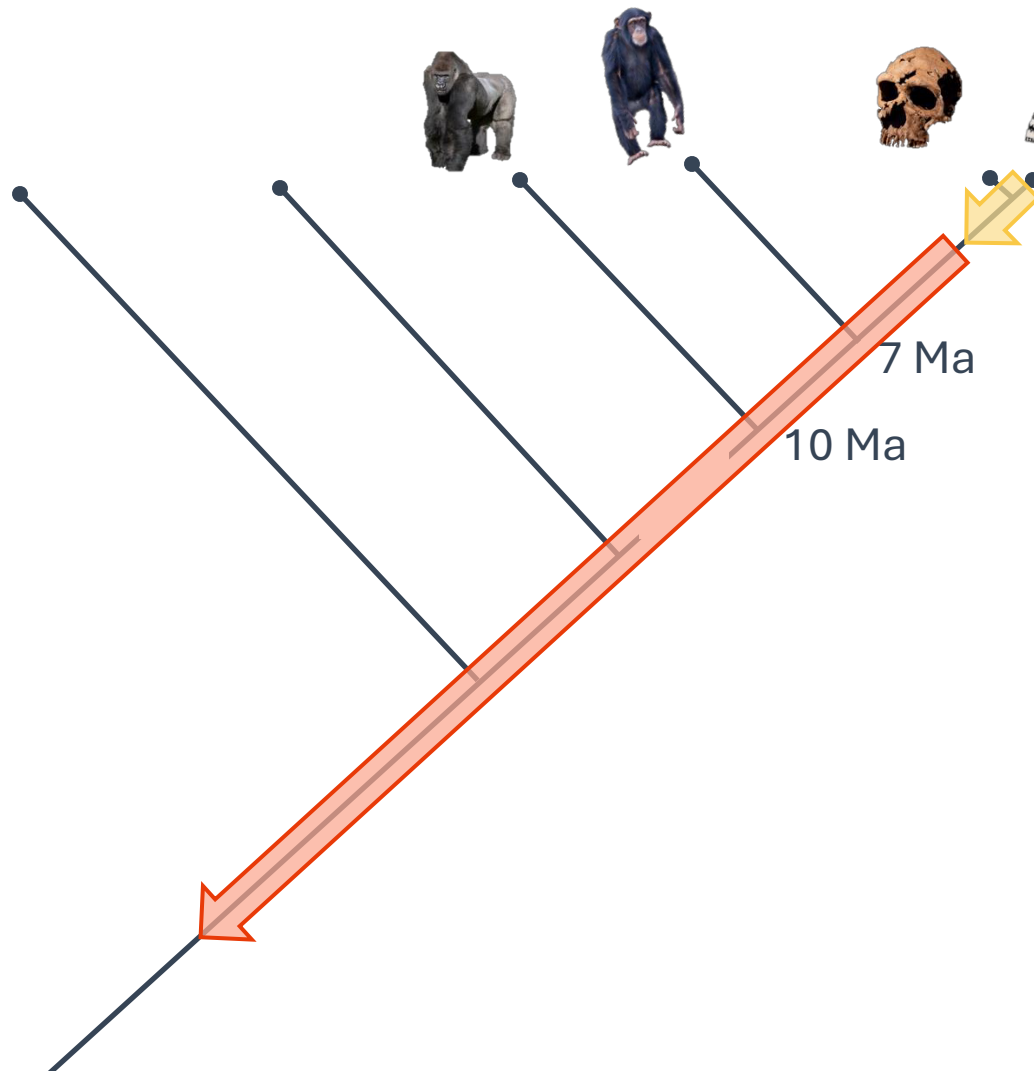
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- **Globular cranium**
- High capacity for plasticity (?)
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set us apart from
Near/ethnals
occurred
after
split
of near/ethnals
evolved
sometime
after
Lmgg

Lectures: when and why did these invariant traits evolve?

Important (and interesting!) overarching question: do you expect there to be a trend in plasticity?

Would you expect more recent traits to be more or less plastic?



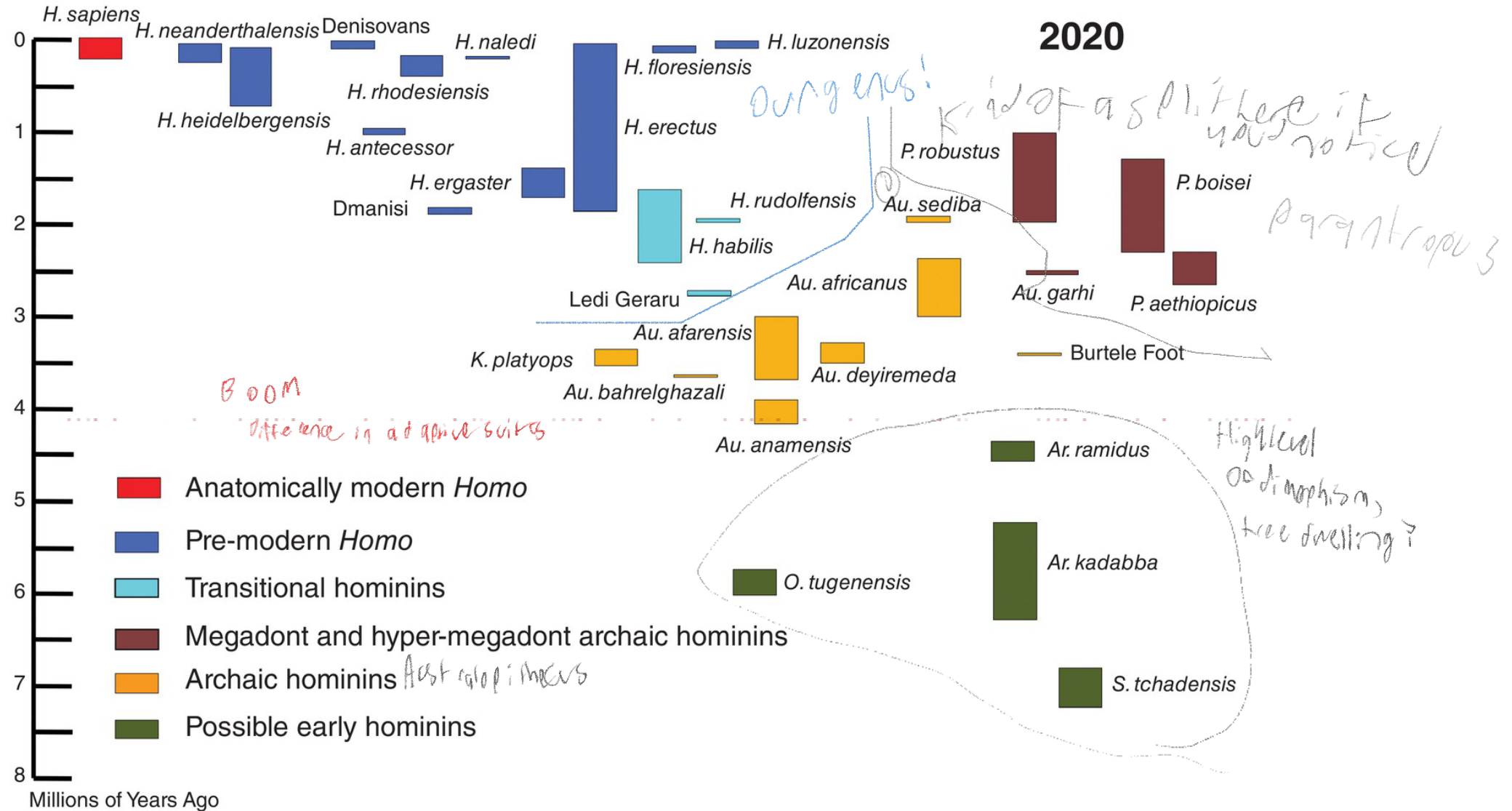
Today: 0-2 Ma

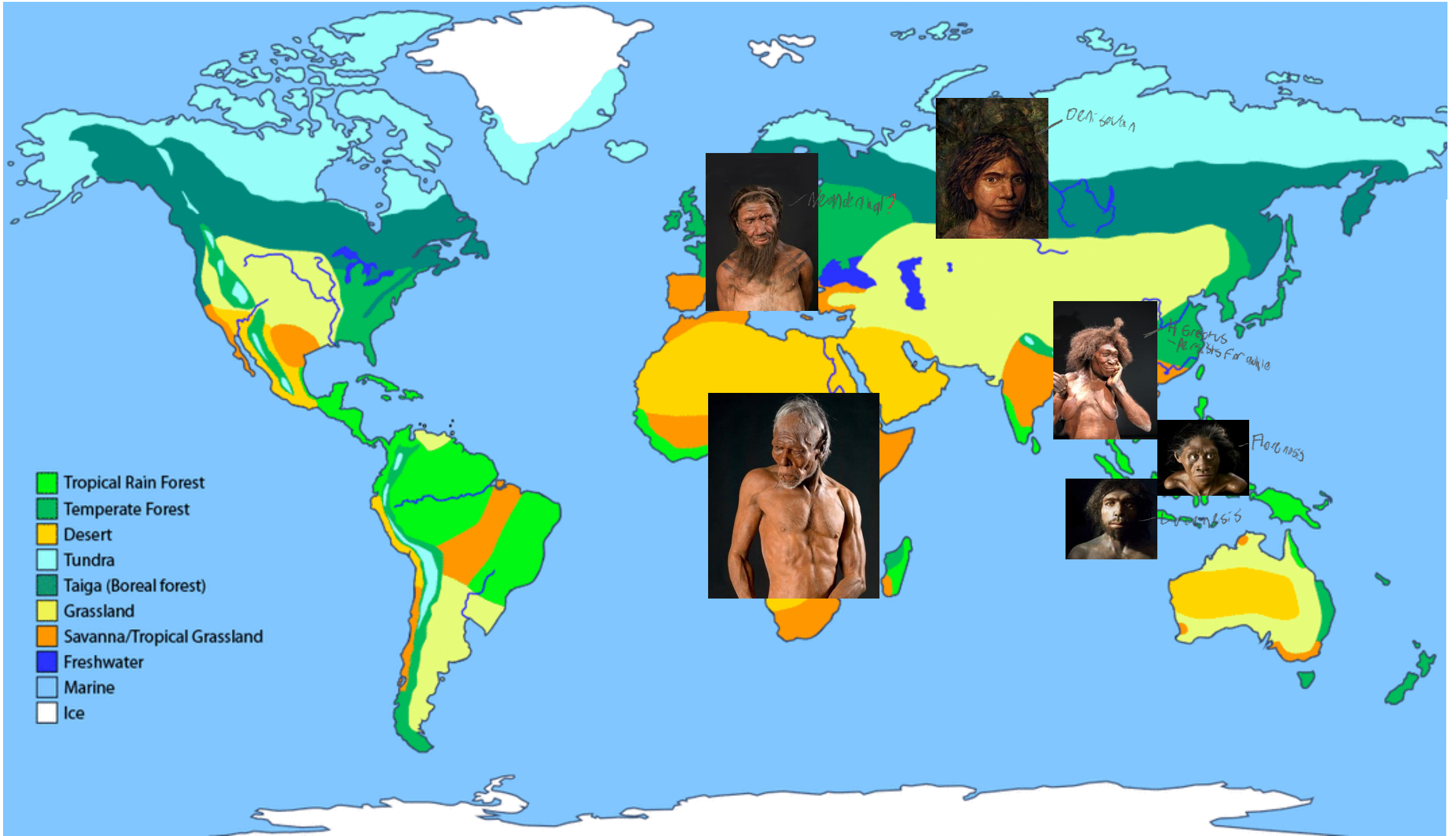
- *Homo sapiens*
- Plasticity and modern human body proportions

Thursday: 2-390 Ma

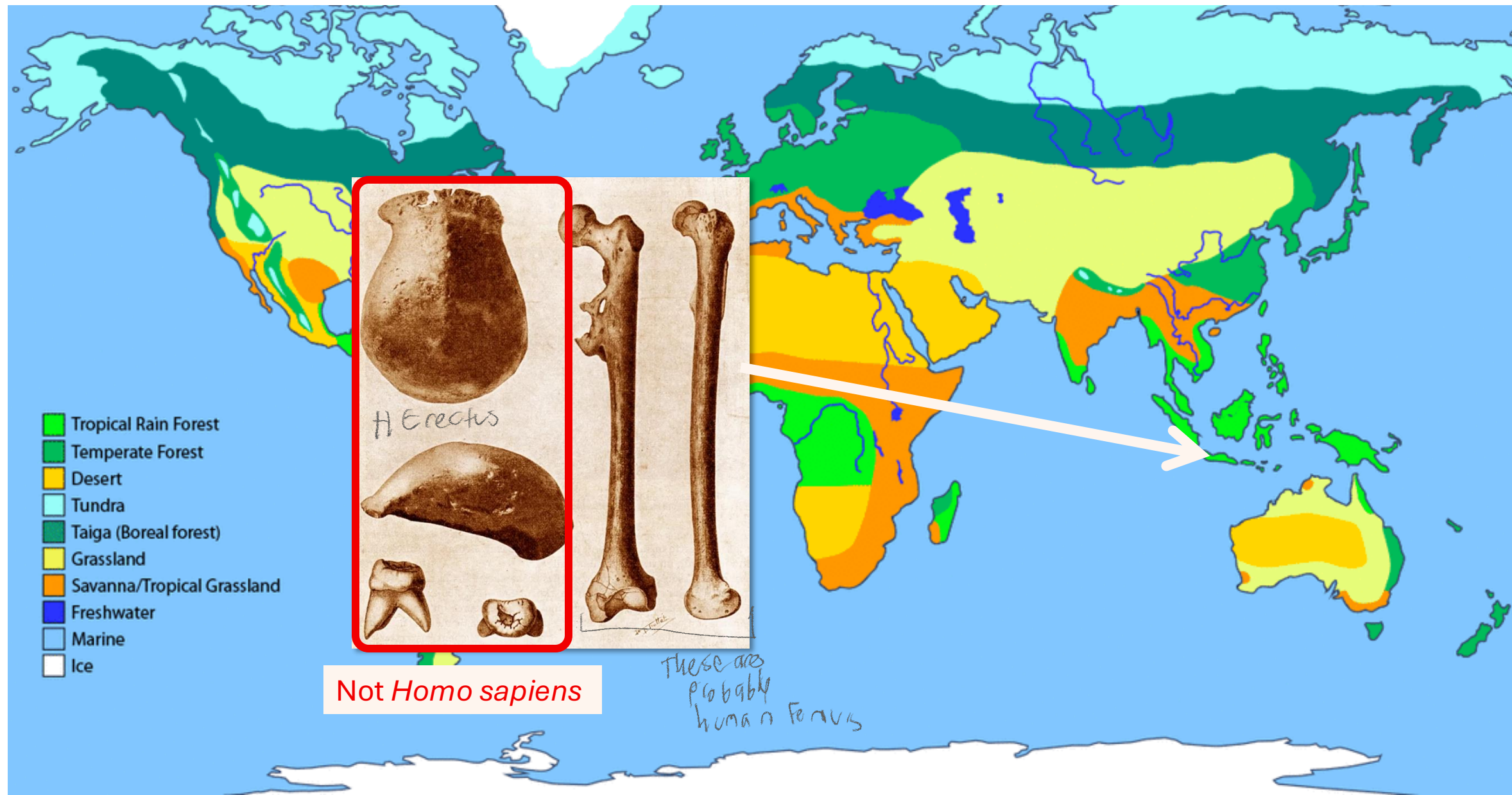
- Body plans: bipedalism, tradeoffs and dilemmas, and throwing capacity
- Ape and anthropoid characters
- Mammalian and vertebrate characters

The quick story





Homo sapiens: a palaeontological definition



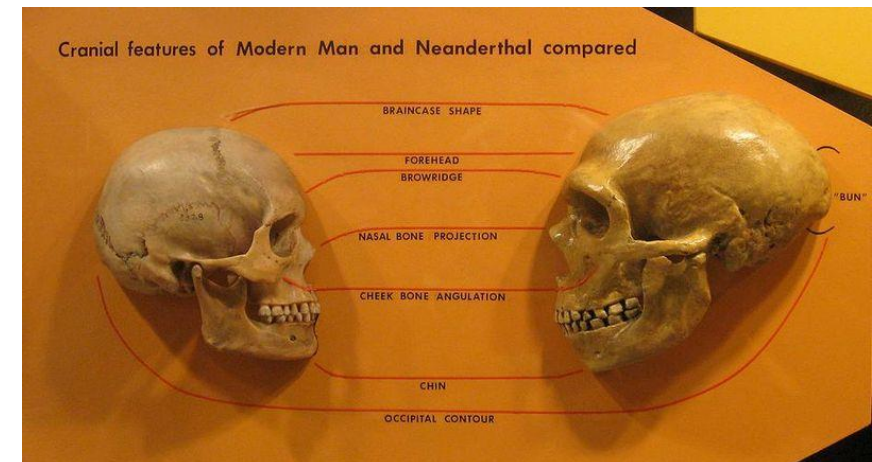
Homo sapiens: a palaeontological definition

Large-brained, bipedal hominin

compared to other hominins -

- Comparatively tall & narrow body
- Globular cranium
 - Small face positioned under the vault
 - Tall forehead
- Comparatively reduced cranial superstructures
 - Reduced occipital bun
 - Reduced browridges
 - Generally gracile
- Mental eminence on mandible

we look like
this due
to adaptations
in
Africa?

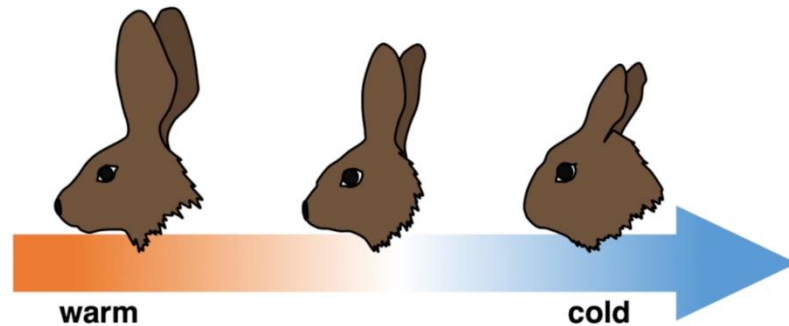


Split from Neanderthals

Homo sapiens: warm-adapted bodies?

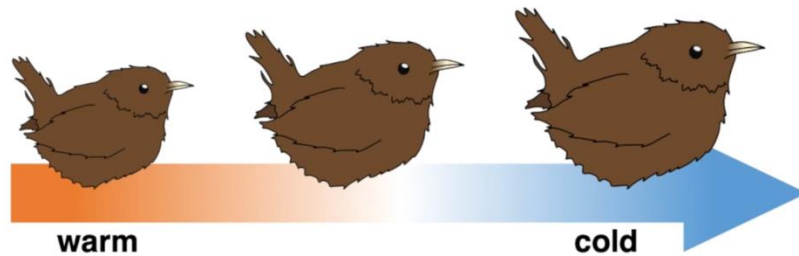
Or in other words: were Neanderthals cold-adapted?

Allen's rule (Allen; limb length!):

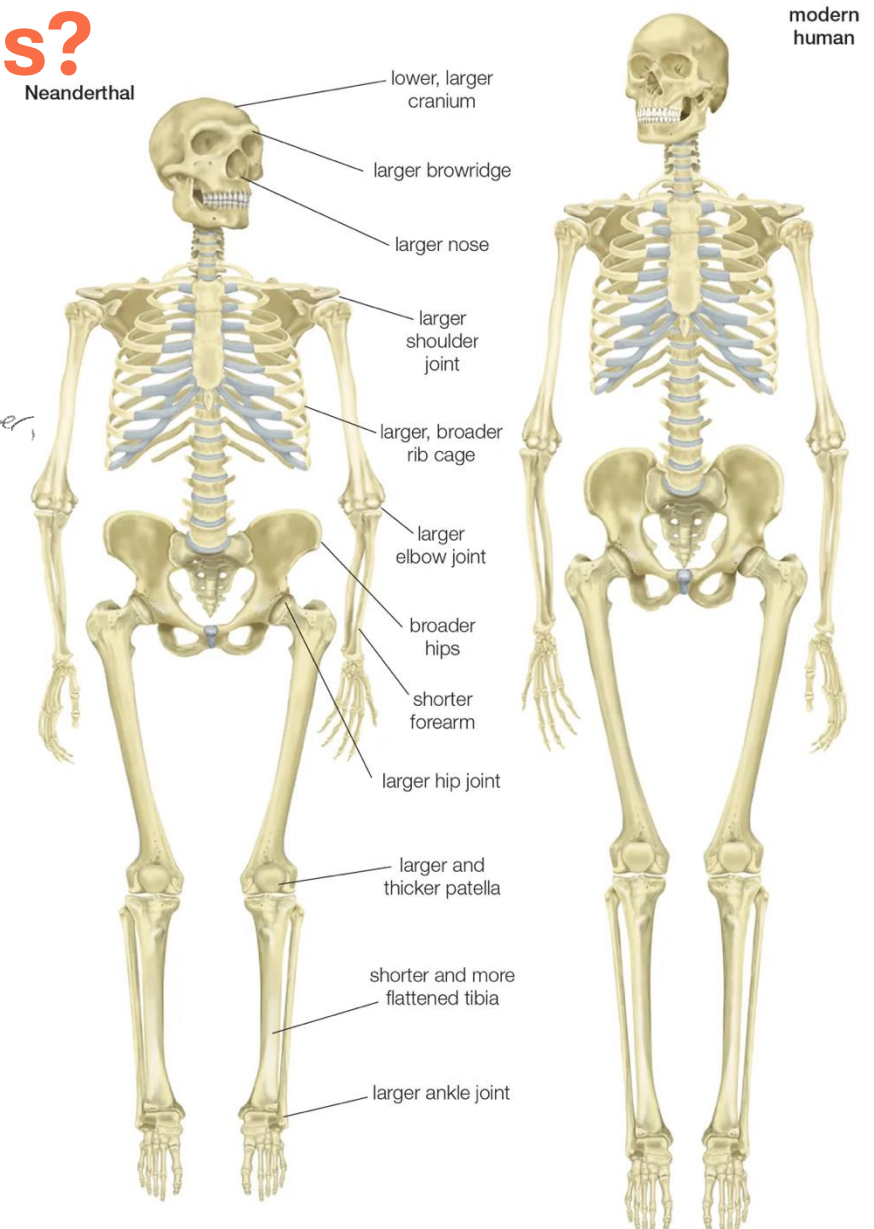


about extremities
warm climates → longer extremities → heat dissipation, adaptation to heat stress
Cold → shorter extremities to retain heat

Bergmann's rule (Bergman; body size):



Cold → bigger bodies to retain heat
Hot → more surface area vs volume for heat dissipation

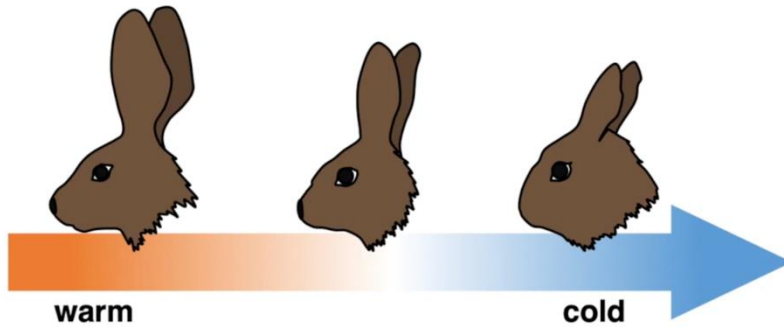


Split from Neanderthals

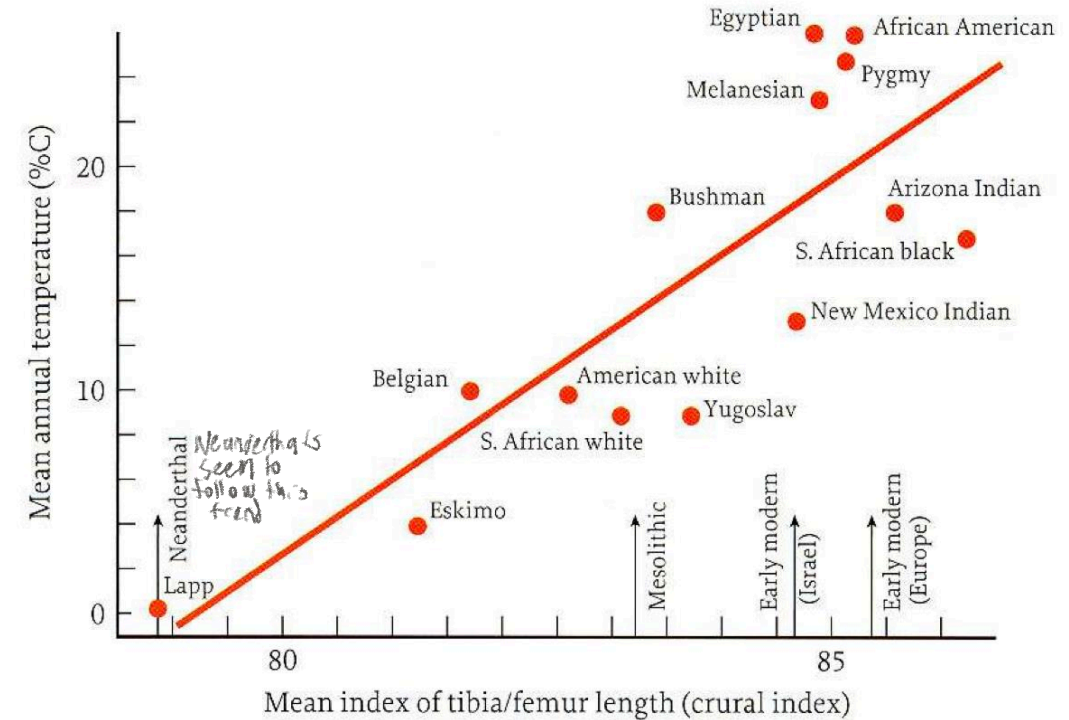
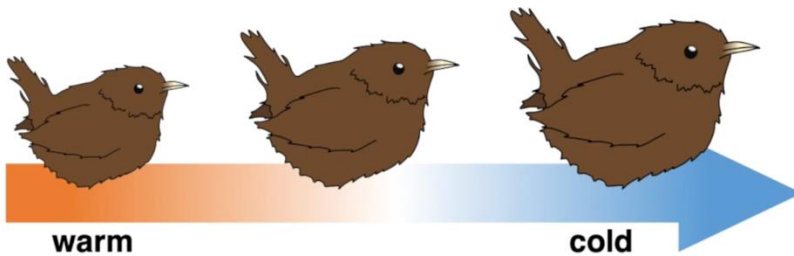
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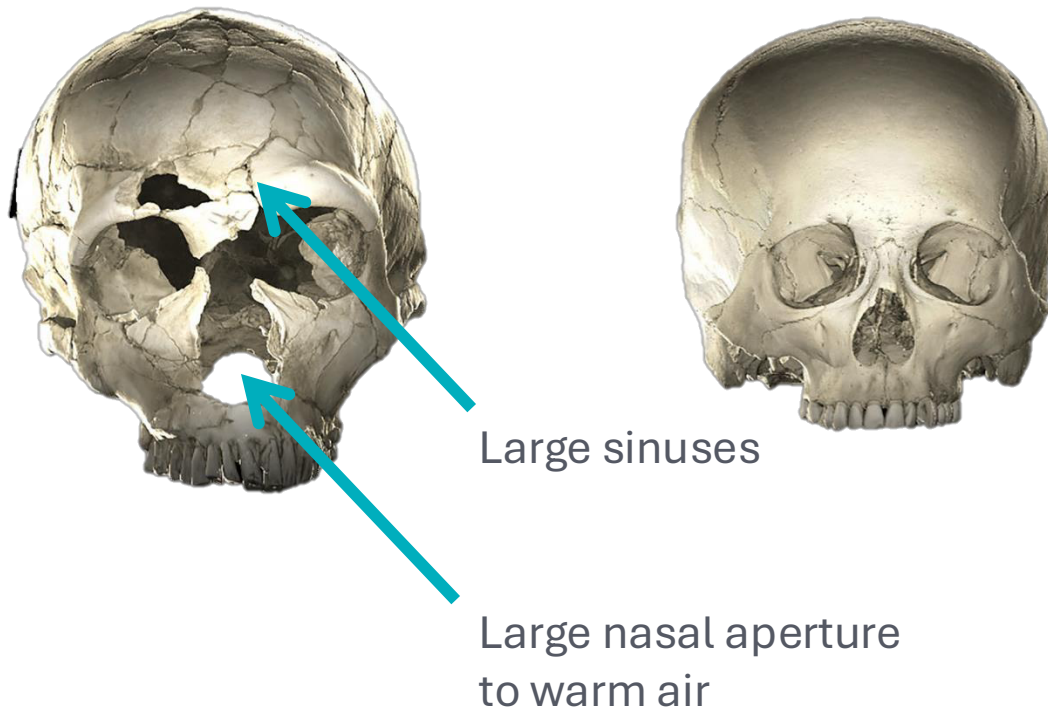


Bergmann's rule (Bergman; body size):



Homo sapiens: warm-adapted bodies?

Or in other words: were Neanderthals cold-adapted?



Split from Neanderthals

2 Ma

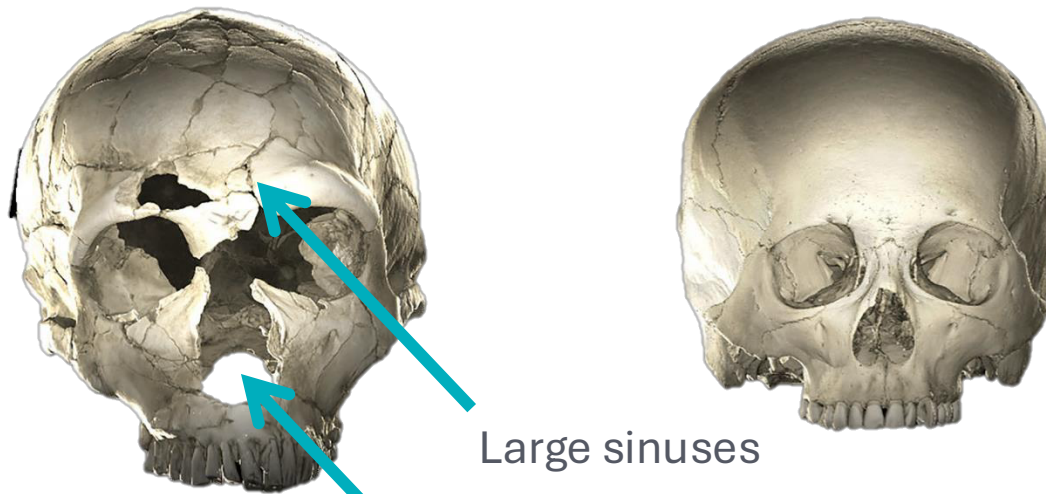
1 Ma

0.5 Ma

Homo sapiens: warm-adapted bodies?

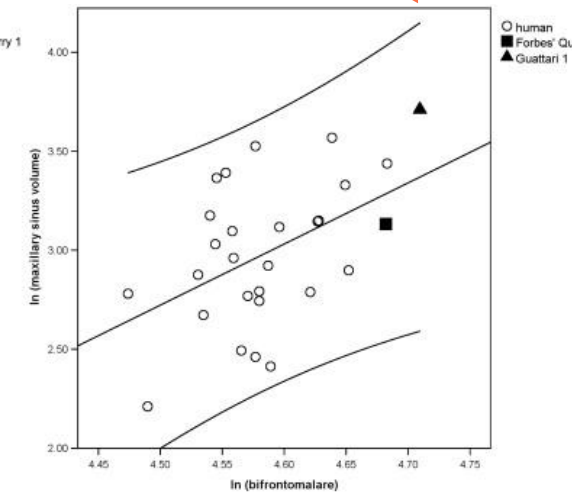
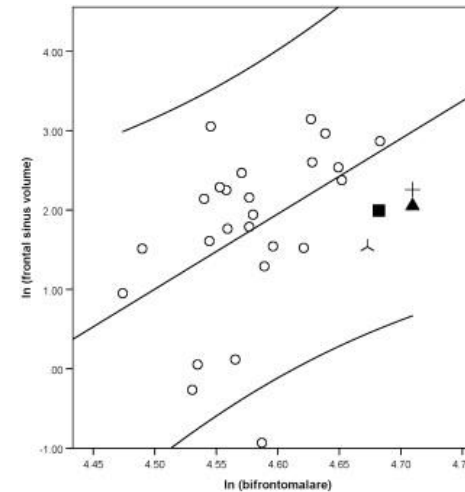
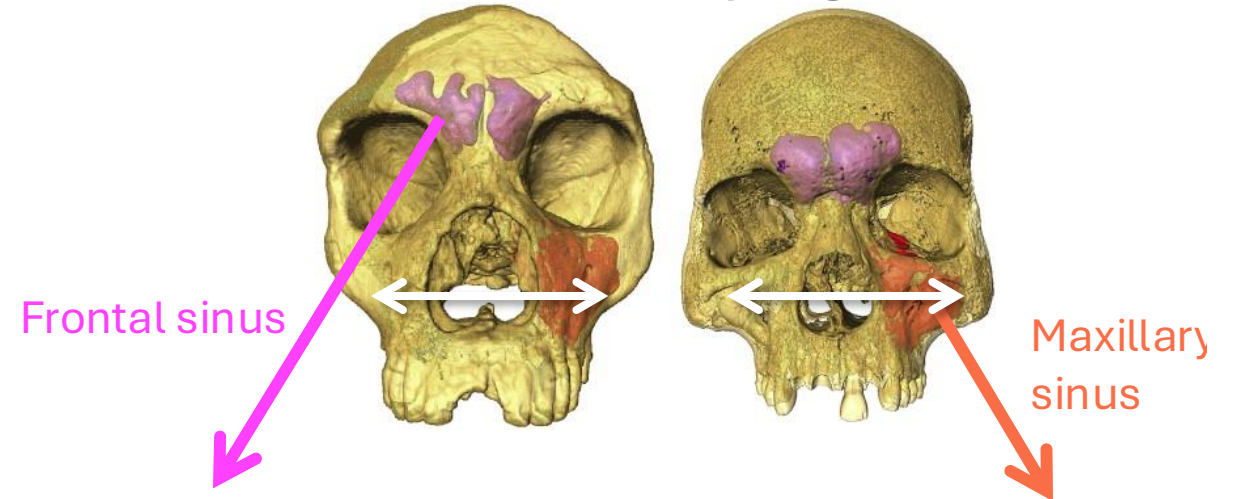
Or in other words: were Neanderthals cold-adapted?

Were Neanderthal sinuses *actually* large – for their faces?



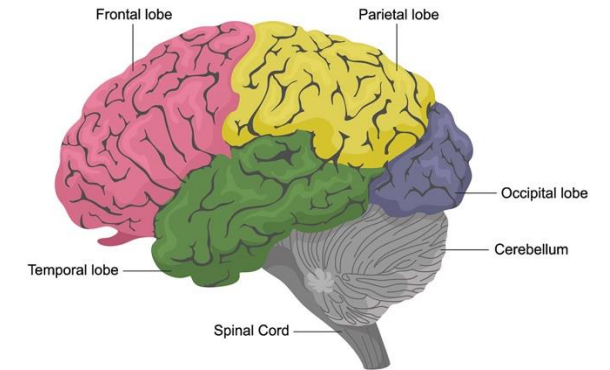
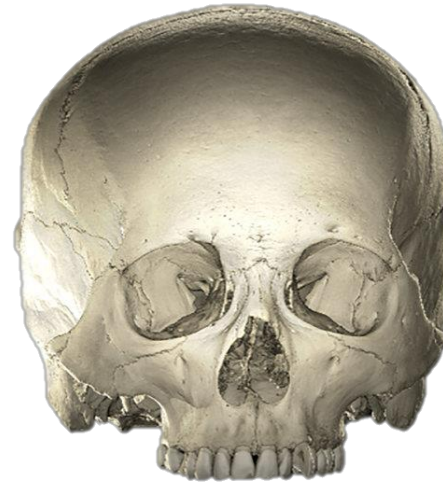
Large sinuses

Large nasal aperture
to warm air

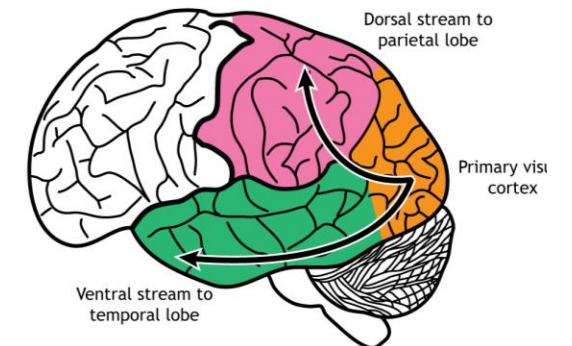


Split from Neanderthals

Homo sapiens: smaller visual cortex?



Visual processing:



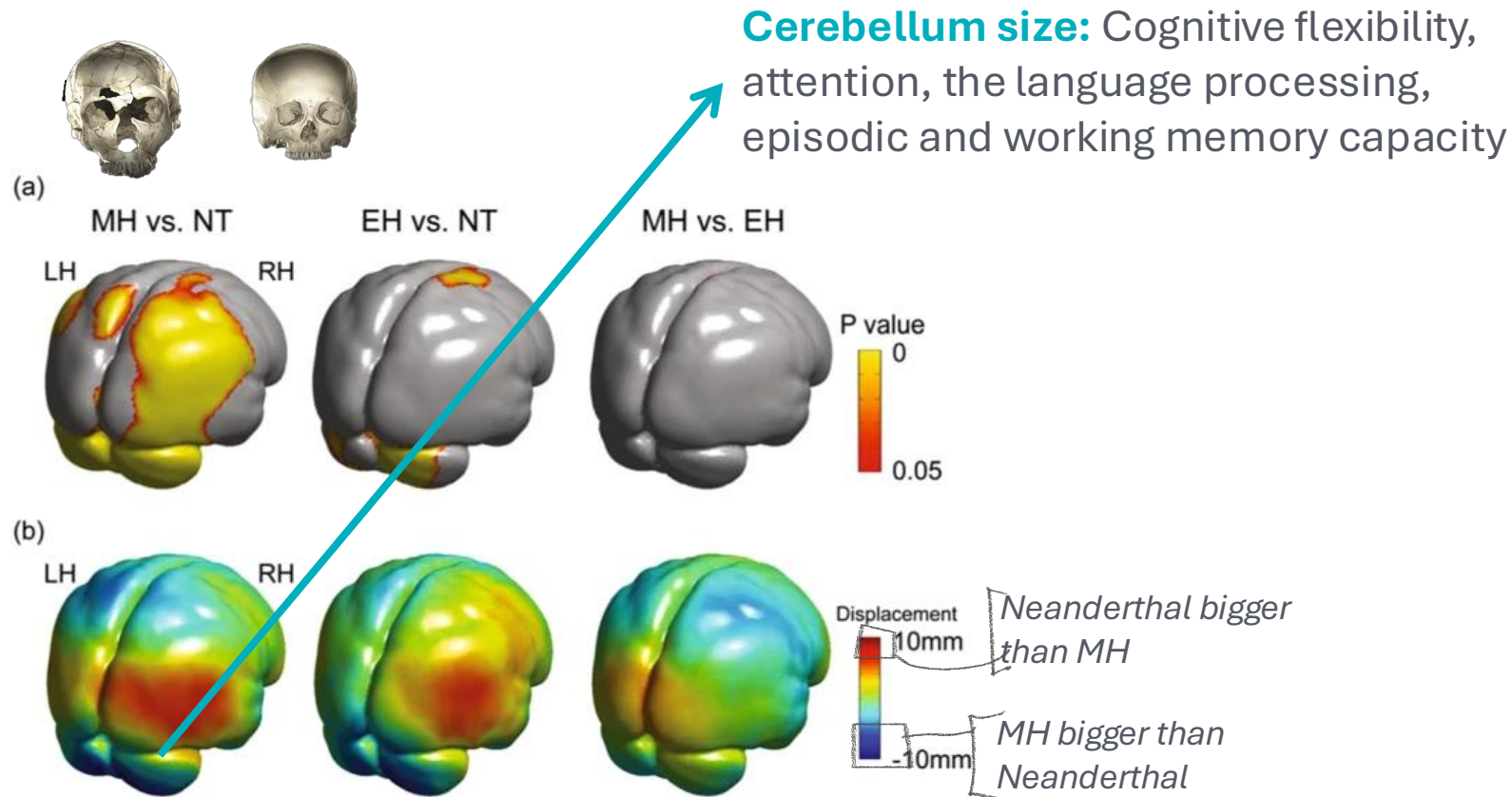
Split from Neanderthals

2 Ma

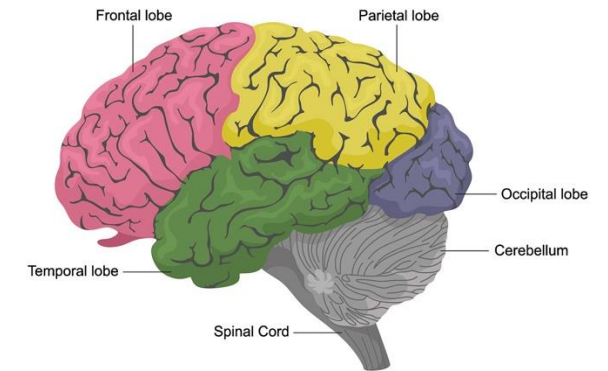
1 Ma

0.5 Ma

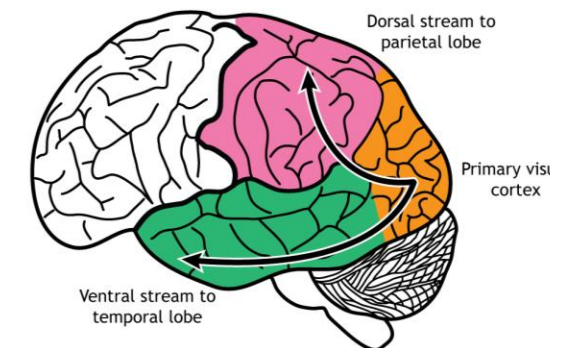
Homo sapiens: smaller visual cortex, larger cerebellum



Neanderthals and modern humans: Differences in **cerebellar, parietal, occipital and medial temporal** regions, but no differences in the frontal regions



Visual processing:



Homo sapiens: origins

~ 200 kya

Skhul V (90 kya) - from Levant

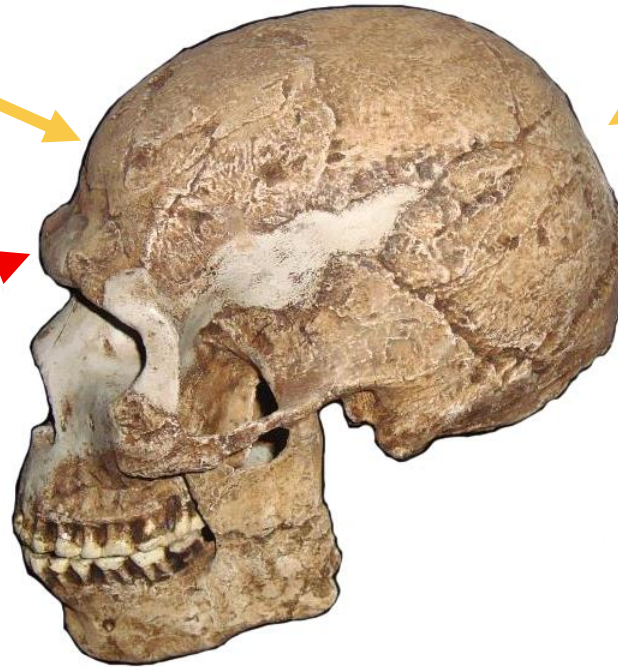
Robust, has Neanderthal traits but is a human - could there be interbreeding?

High, flat forehead
- Human

Rounded braincase
- Human

Projecting brow ridge
- Neanderthal

Occipital bun
- Neanderthal



0.1 Ma

Split from Neanderthals



Homo sapiens: origins

~200 kya
Qafzeh (90-120 kya) - Also from Leant

High, flat forehead

Rounded braincase

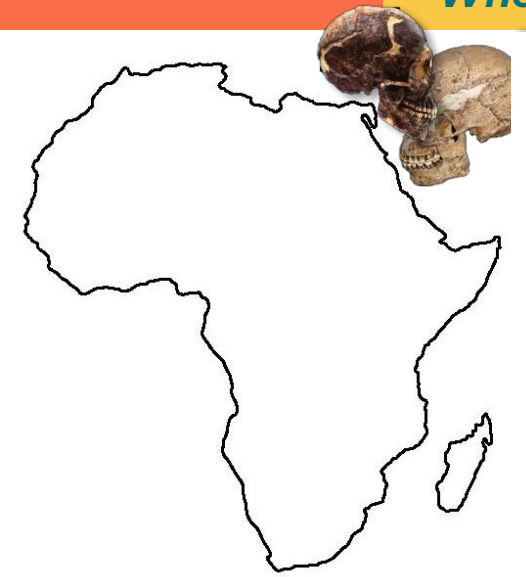
still kind of archaic

Projecting brow ridge, but within modern human range



Qafzeh 9

Mental eminence



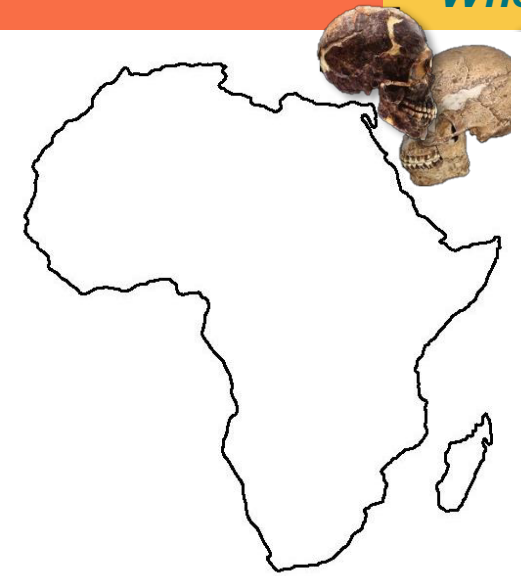
0.1 Ma

Split from Neanderthals



Homo sapiens: origins

~100 kya mosaics of traits in Cranium - some
Modern, some archaic



0.2 Ma Mosaics;
brow ridges
not yet flat



Split from Neanderthals

0.5 Ma

1 Ma

2 Ma

Homo sapiens: origins

Context

Homo sapiens

Plasticity

Body plans



Herto (160 kya) — From Ethiopia

Similar to Skhul
More archaic looking
Homo sapiens from 100 kya



Herto

Globular cranium

Kinda archaic

Projecting brow
ridge – but within
modern human
range



Occipital bun

Archaic

Very robust, very large

Archaic



0.2 Ma

Mosaics;
brow ridges
not yet flat

Split from Neanderthals

0.5 Ma

2 Ma

1 Ma

Homo sapiens: origins

Context

Homo sapiens

Body plans, plasticity

Omo (233 kya)

Omo I relatively modern-looking

kind of archaic

Projecting brow ridge – but within modern human range

Globular cranium

Mental eminence



0.2 Ma

Mosaics; brow ridges not yet flat

Split from Neanderthals



2 Ma

1 Ma

0.5 Ma

Homo sapiens: origins



Mosaics;
brow ridges
not yet flat

Split from Neanderthals



Context

Homo sapiens

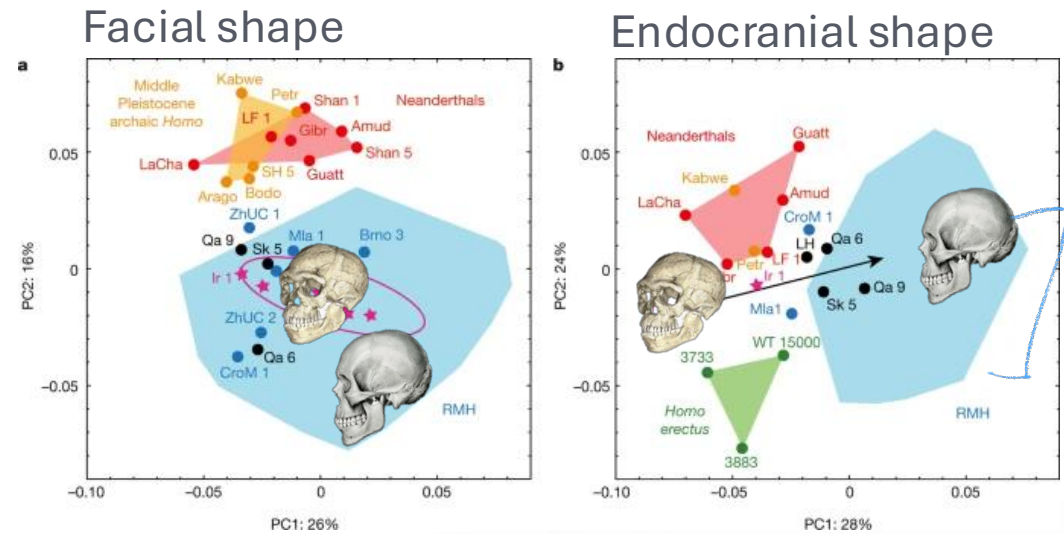
Body plans, plasticity

2 Ma

1 Ma

0.5 Ma

Homo sapiens: origins



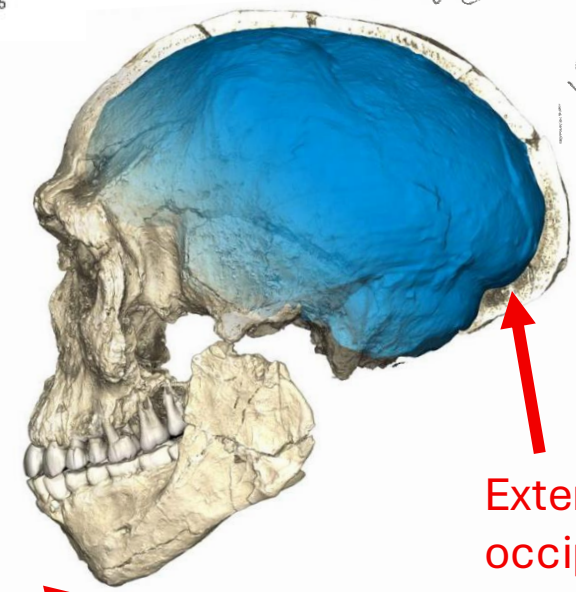
First unequivocal human example as of now
Jebel Irhoud (315 kya)
Overall shape is what is modern

Archaic traits...

Projecting brow ridges



No chin



Extended occipital



Mosaics; brow ridges not yet flat

0.3 Ma

Split from Neanderthals

2 Ma

1 Ma

0.5 Ma

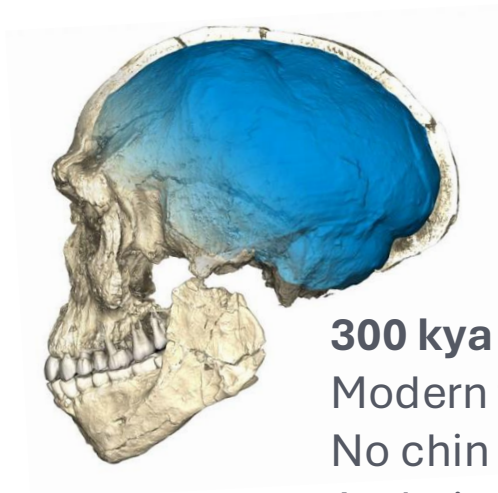
Context

Homo sapiens

Body plans, plasticity

Homo sapiens: origins summary

- Face established relatively early
- Brain keeps evolving



300 kya
Modern face
No chin
Archaic brain



Omo



Herto

200 kya
Modern and archaic faces
Chin
Modern and archaic brain



Qafzeh



Skhul

100 kya
Modern and archaic faces
Chin
Modern and archaic brain



Split from Neanderthals

Homo sapiens: origins summary

- Face established relatively early
- Brain keeps evolving
- **DIVERSITY**

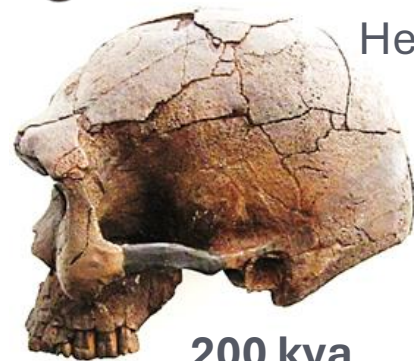
Humans morphologically diverse



Omo



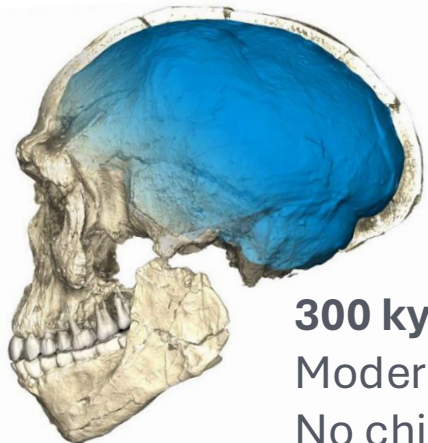
Qafzeh



Herto



Skhul



300 kya

Modern face

No chin

Archaic brain

200 kya

Modern and archaic faces

Chin

Modern and archaic brain

100 kya

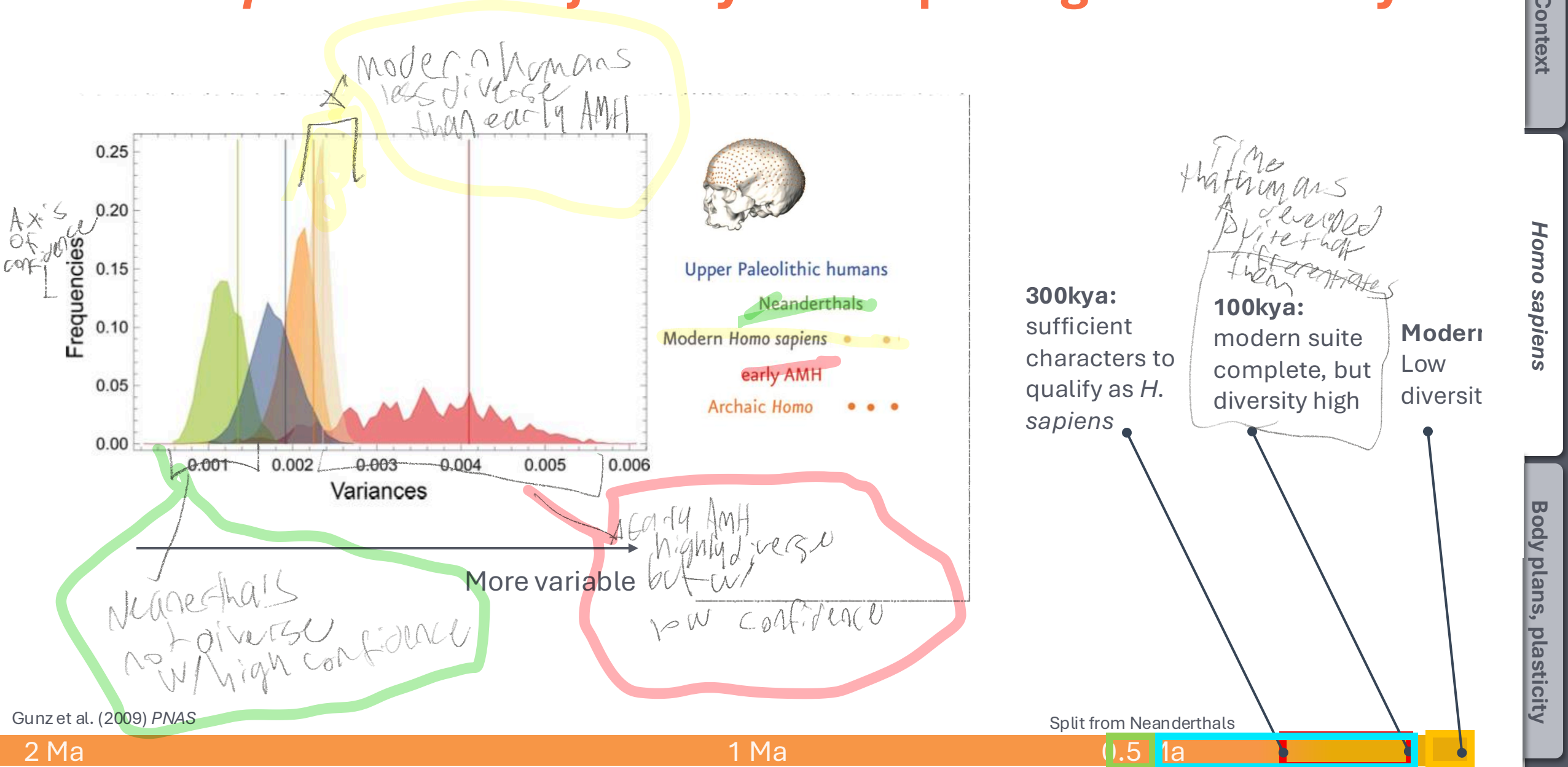
Modern and archaic faces

Chin

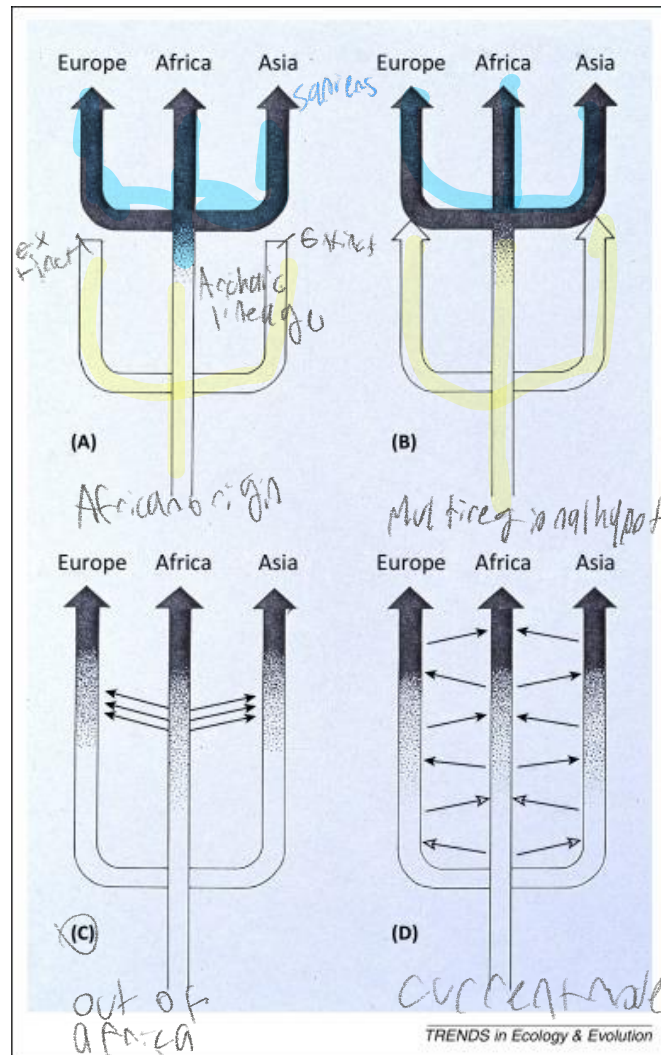
Modern and archaic brain

Split from Neanderthals

Homo sapiens: the trajectory of morphological diversity



Homo sapiens: evolution of the species



C - single ancestral population, admixture w/ archaic lineage

Split from Neanderthals

2 Ma

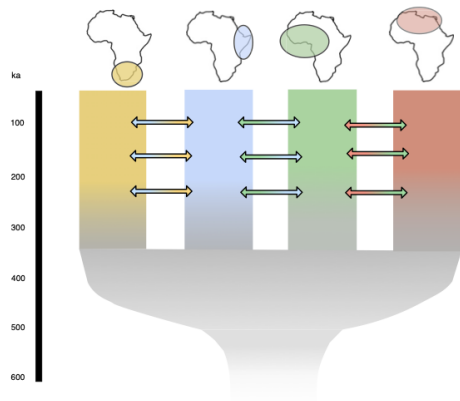
1 Ma

0.5 Ma

w/ interbreeding of archaic humans

Homo sapiens: evolution of the species

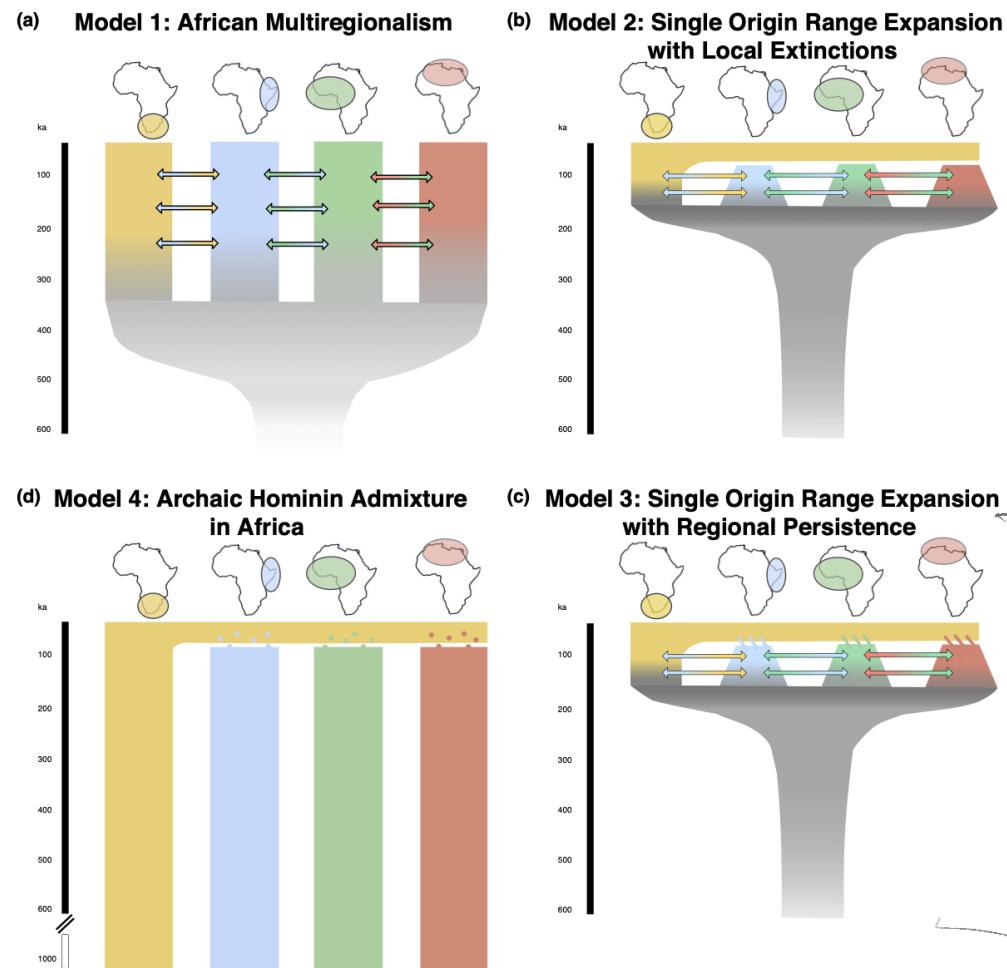
(a) Model 1: African Multiregionalism



Sapiens arise from
different pops in Africa
coalescing instead
of single African pop



Homo sapiens: evolution of the species



Current Opinion in Genetics & Development

Different
variations
on multiple
source populations
Africa

Most
likely
model
now

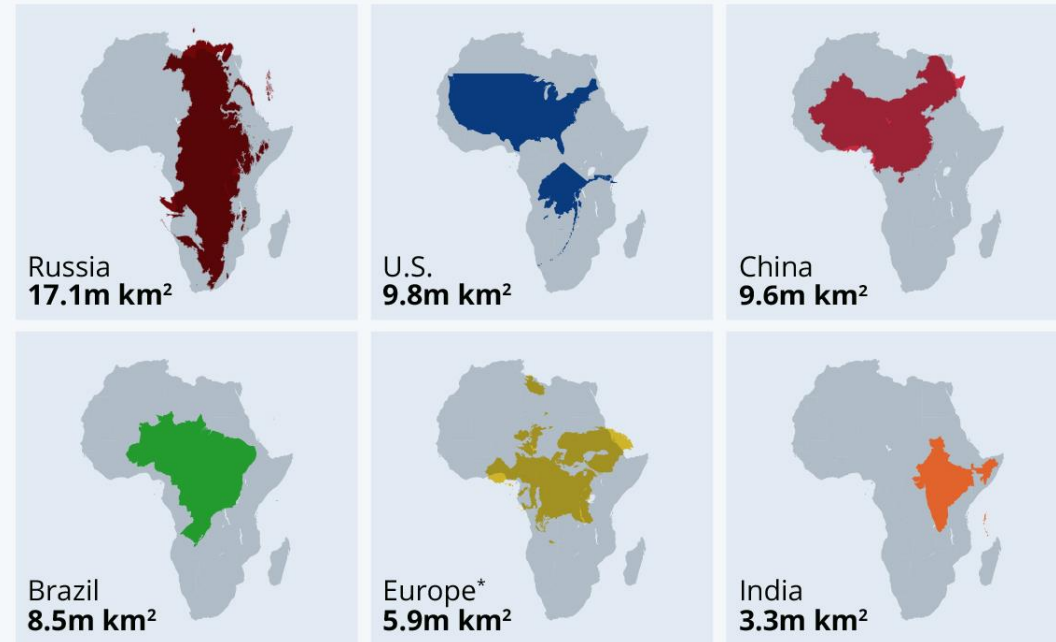


Homo sapiens: evolution of the species

The True Size of Africa

Comparison of Africa's land area to selected countries/regions

African continent **30.4m km²**



* excluding Russia.

Sources: CIA, World Population Review, Visual Capitalist



statista

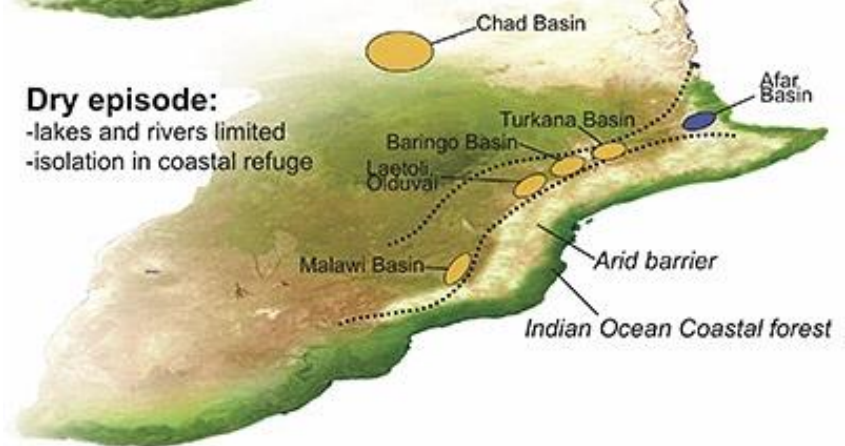
Wet episode:

- lakes and rivers more prominent
- dispersal via riverine corridors



Dry episode:

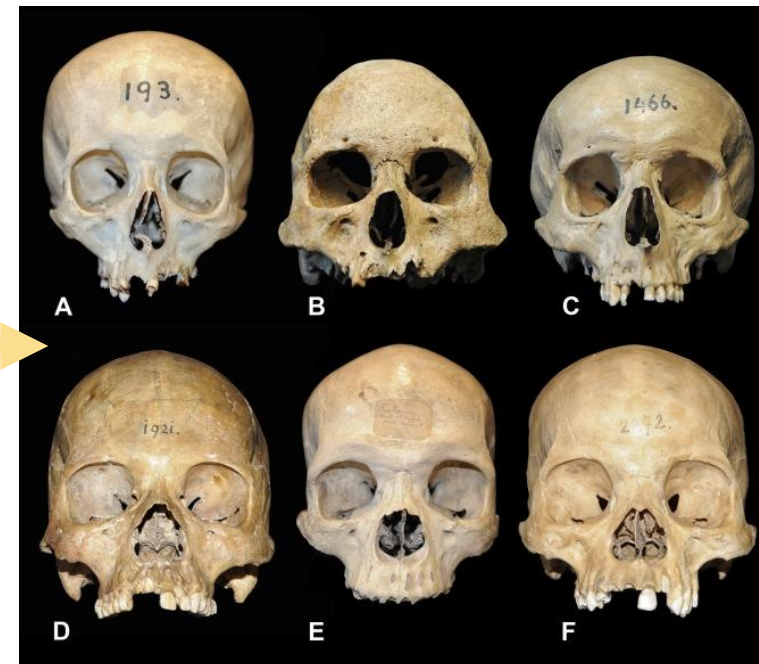
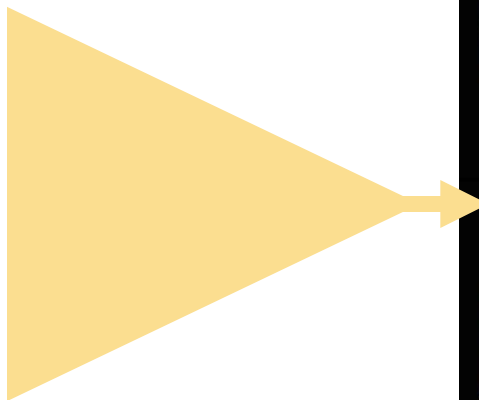
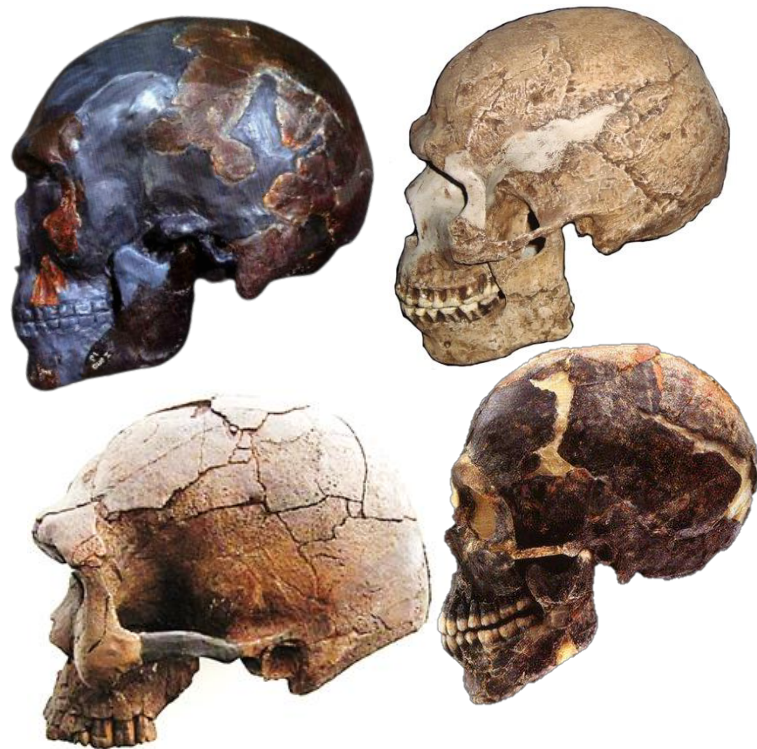
- lakes and rivers limited
- isolation in coastal refuge



Split from Neanderthals

Homo sapiens: why a reduction in diversity?

Or in other words: what happens after the establishment of the modern human package?



Split from Neanderthals

2 Ma

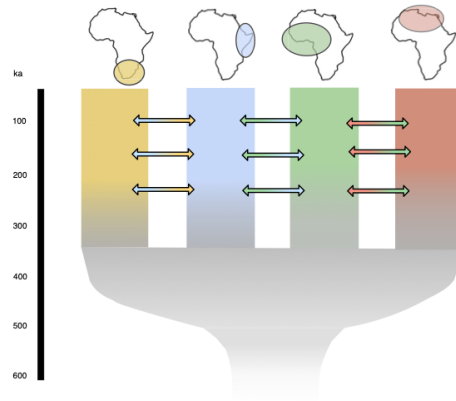
1 Ma

0.5 Ma

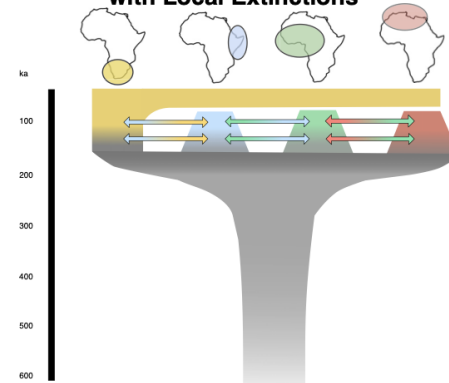
Homo sapiens: why a reduction in diversity?

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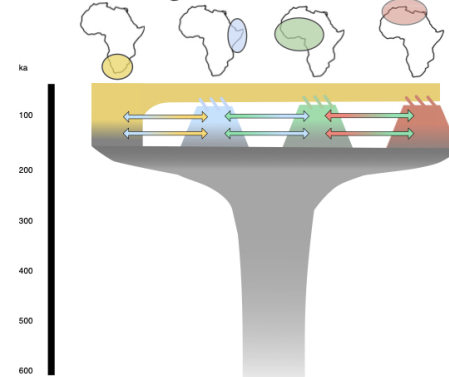
(a) **Model 1: African Multiregionalism**



(b) **Model 2: Single Origin Range Expansion with Local Extinctions**

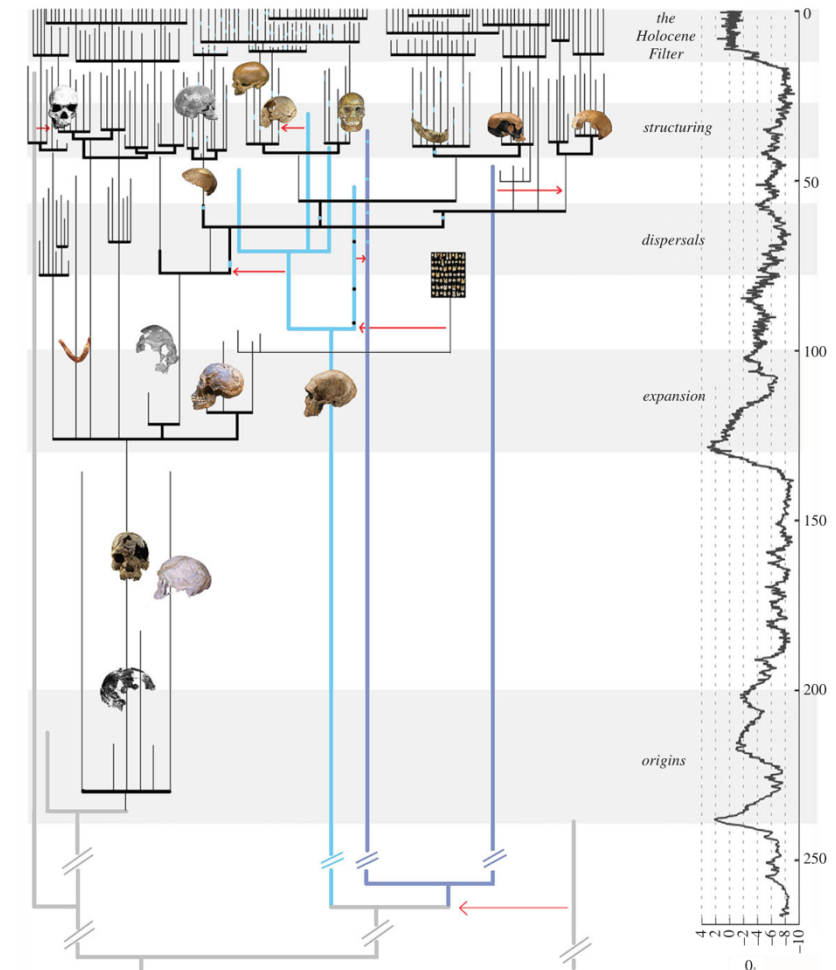


(c) **Model 3: Single Origin Range Expansion with Regional Persistence**



Current Opinion in Genetics & Development

If a), strong shared selection for reduced diversity

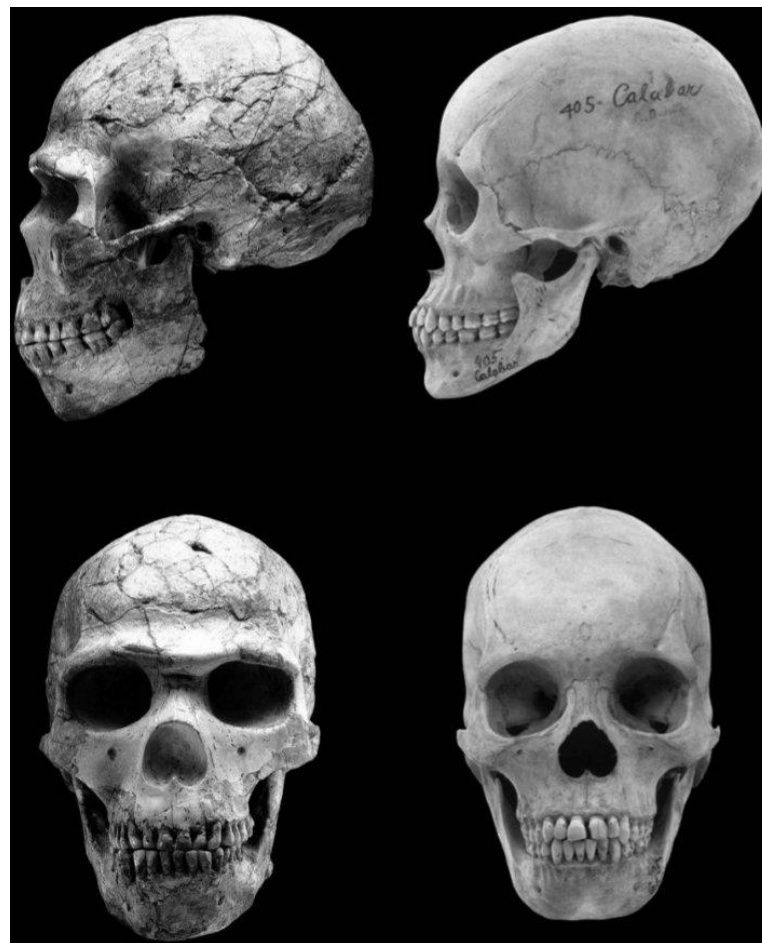


Mirazon Lahr. (2016)

Split from Neanderthals

Homo sapiens: why a reduction in diversity and robusticity?

Or in other words: what happens after the establishment of the modern human package?



1. Self-domestication (?)
2. Neolithic transition

Split from Neanderthals

Domestication

Farm fox experiment
(e.g., Trut et al. 1999)

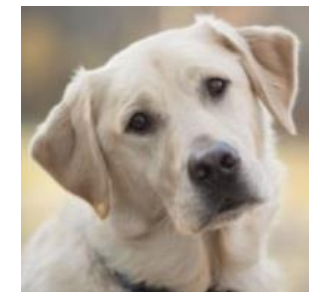
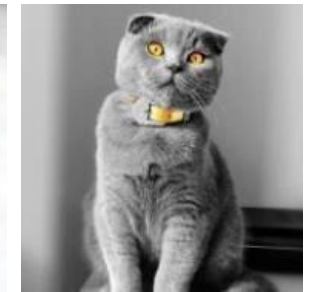


Domestication syndrome...

- Floppy ears
- Depigmentation
- Reduced brain size; globular brain
- Curly tail
- Gracile facial skeleton
- Increased fecundity
(decreased seasonality, increased litter size, earlier sexual maturity)
- Neotenous (juvenile) behaviour

...as a result of
selection for
tameness/reduced
aggression?

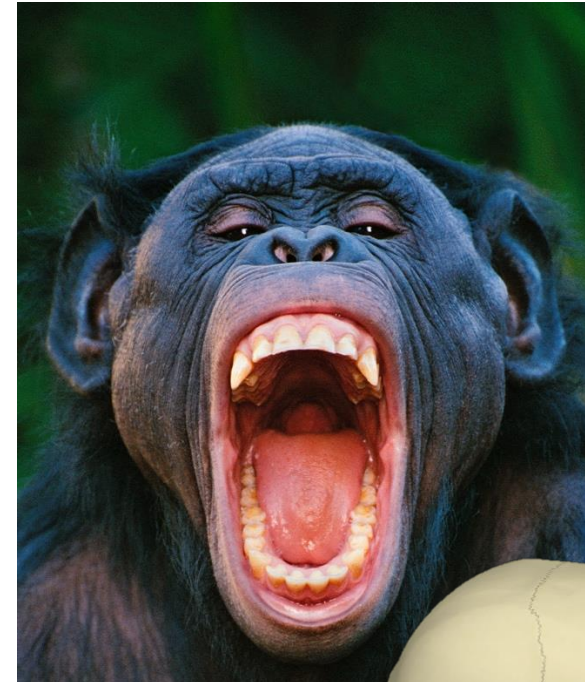
(e.g. Wilkins et al. 2014)



Split from Neanderthals

Self-domestication

Domestication in the absence of interventions by another species



- Chimpanzees hugely more aggressive than bonobos
- Bonobo morphology essentially paedomorphic ^{juvenile-like} chimpanzee (reduced cranial capacity, reduced facial superstructures, reduced canines)
- Despite very similar ecology

Bonobos are example of self-domestication

Hare et al. (2012) *Anim. Behav.*

Split from Neanderthals

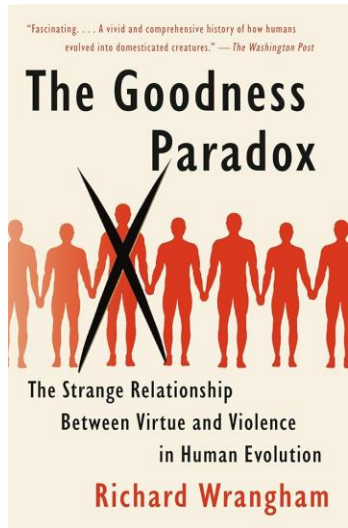
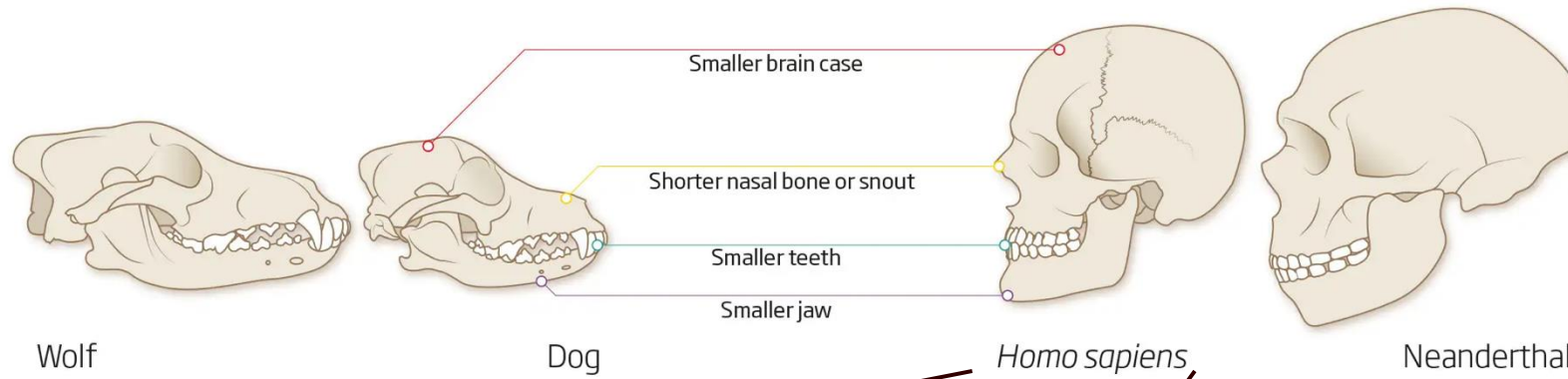
2 Ma

1 Ma

0.5 Ma

Self-domestication

Domestication in the absence of interventions by another species



Selection against *reactive* aggression... through *proactive* aggression?

Was aggression selected out of killing people?
Hard to lose interest in freedom



Domestication syndrome present...
Selection for reduced aggression?

↑ selective pressures for reduced aggression?

Split from Neanderthals

2 Ma

1 Ma

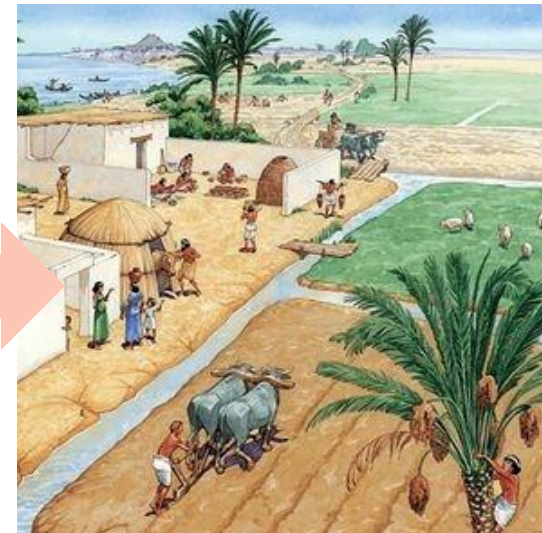
0.5 Ma

Neolithic transition

Hunter-gatherers → Farmers



High activity levels
Diverse, hard food



Low activity levels
Low quality, soft food
Disease

Split from Neanderthals

2 Ma

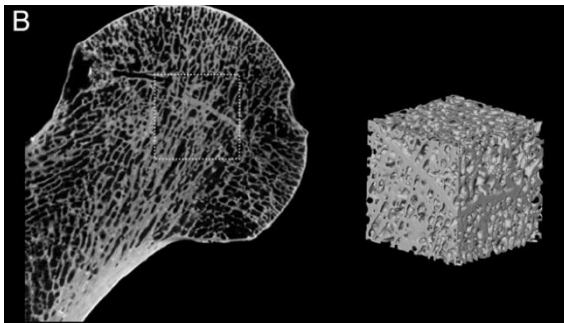
1 Ma

0.5 Ma

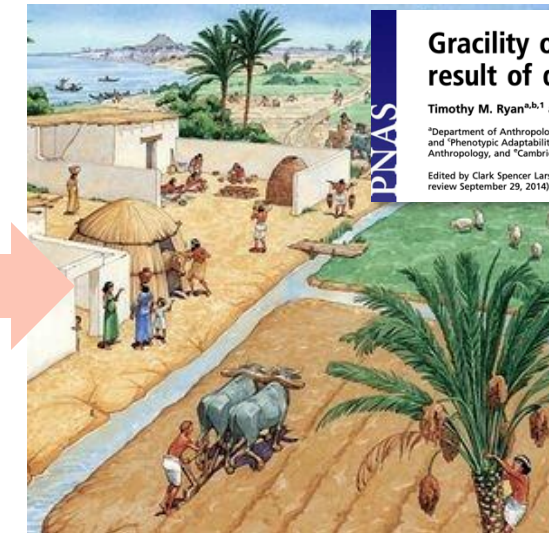
Neolithic transition: effects on morphology



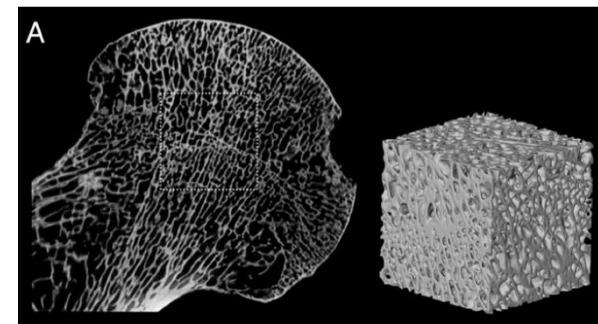
High activity levels
Diverse, hard food



2 Ma



Low activity levels
Low quality, soft food
Disease



1 Ma

Gracility of the modern *Homo sapiens* skeleton is the result of decreased biomechanical loading

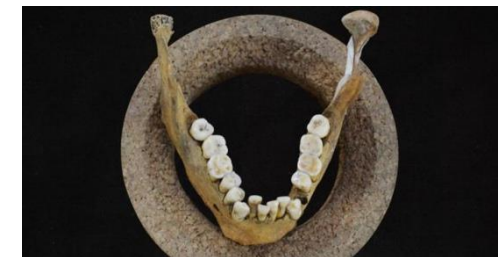
Timothy M. Ryan^{a,b,1} and Colin N. Shaw^{c,d,e}

^aDepartment of Anthropology, ^bCenter for Quantitative Imaging, EMS Energy Institute, The Pennsylvania State University, University Park, PA 16802; and ^cPhenotypic Adaptability, Variation and Evolution Research Group, ^dMcDonald Institute for Archaeological Research, Department of Archaeology and Anthropology, and ^eCambridge BioTomography Centre, Department of Zoology, University of Cambridge, Cambridge CB2 3QG, United Kingdom

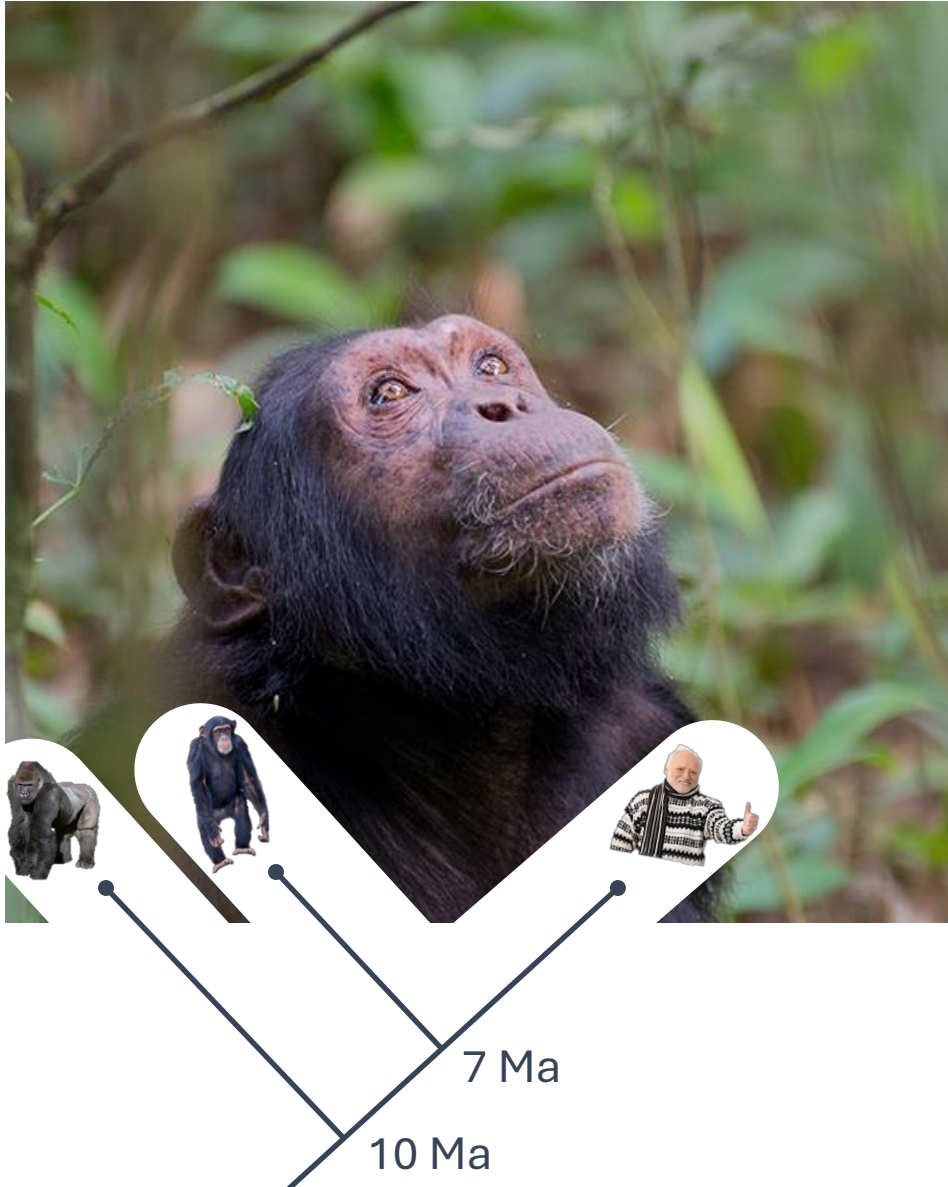
Edited by Clark Spencer Larsen, The Ohio State University, Columbus, OH, and accepted by the Editorial Board November 20, 2014 (received for review September 29, 2014)

Split from Neanderthals

0.5 Ma

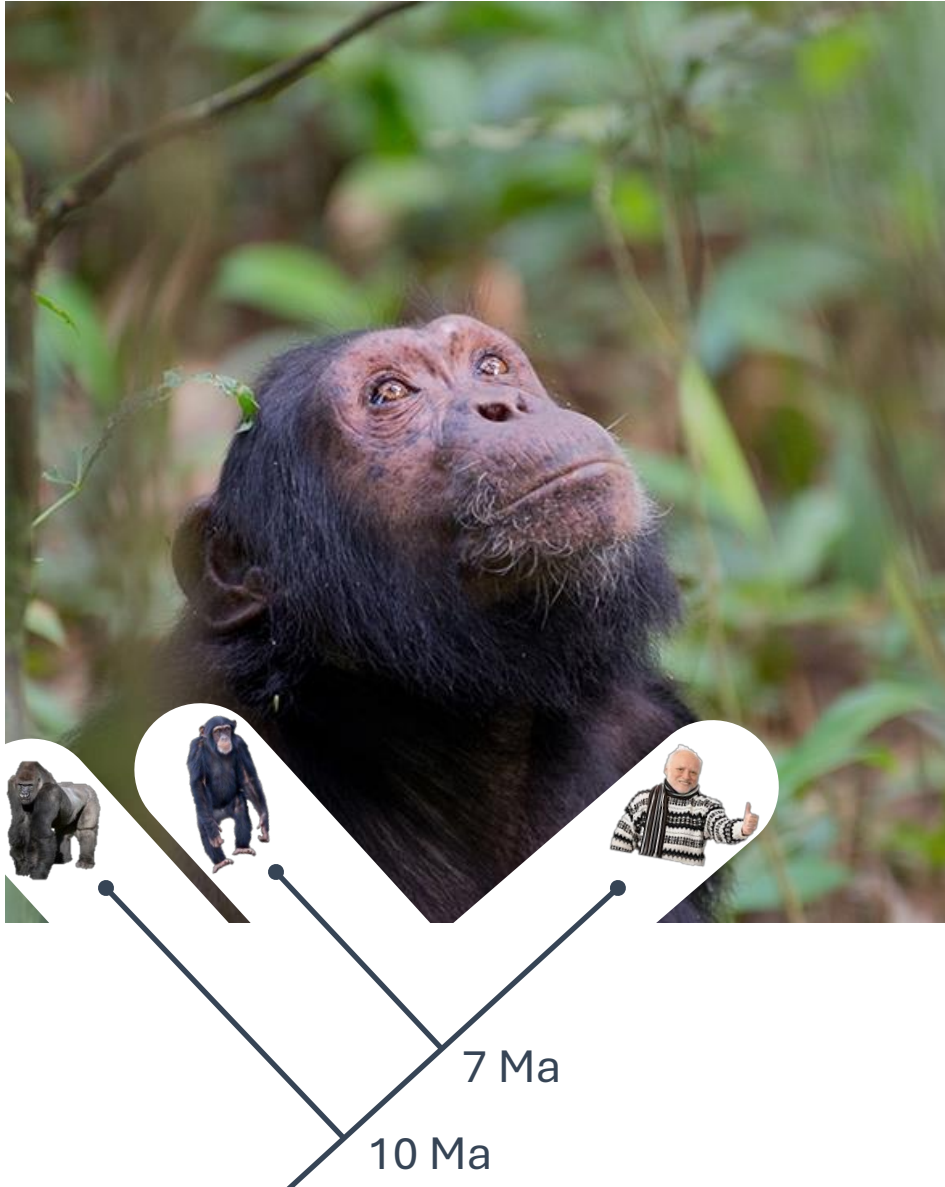


Traits that 'make' us human



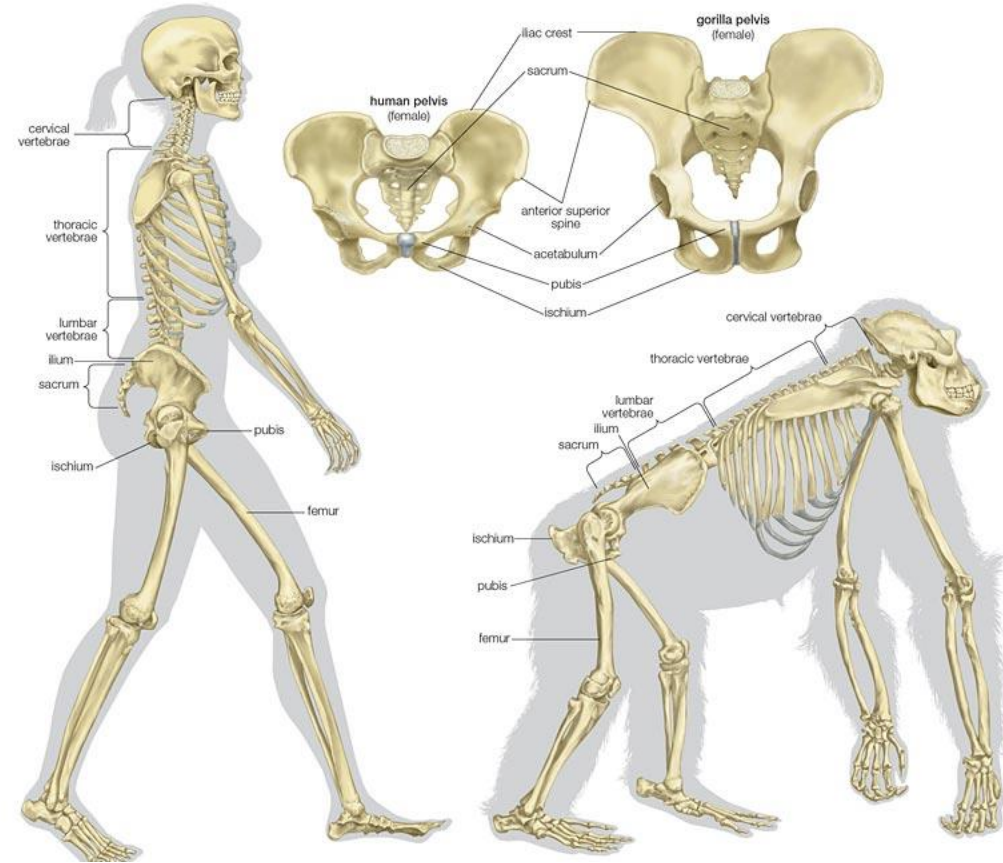
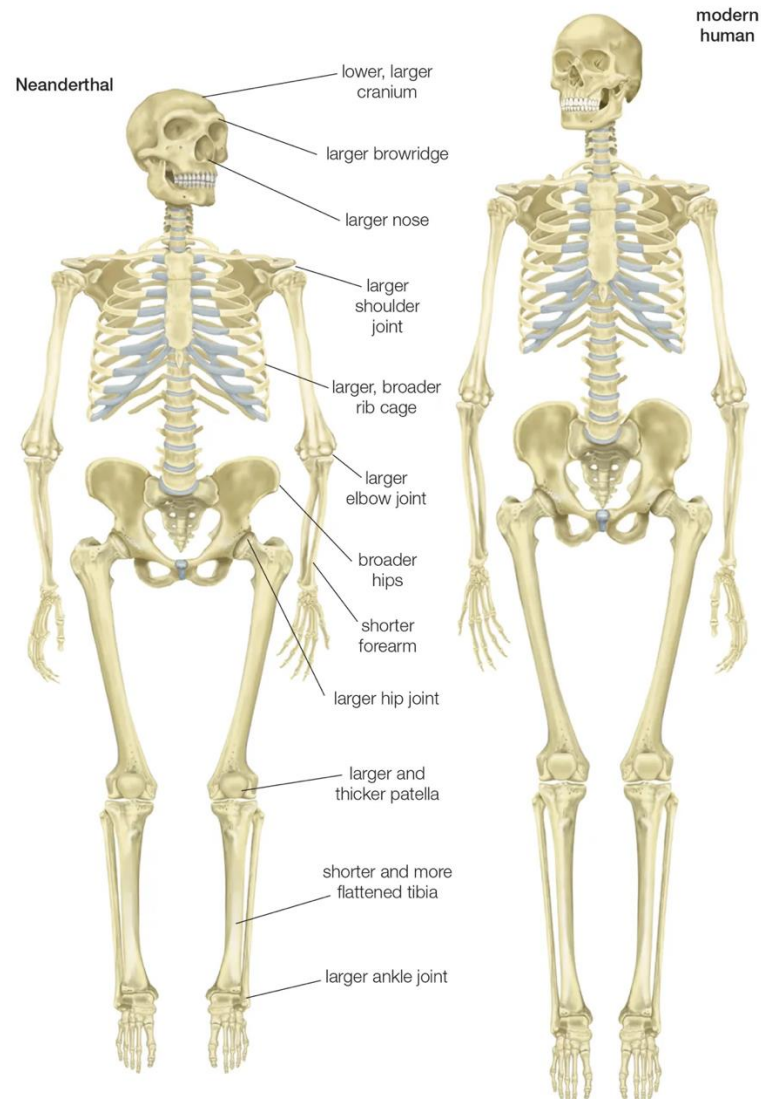
- Obligate terrestrial bipedalism
 - Bowl-shaped pelvis
 - Elongated femur
 - Flexible lower back
 - Highly derived feet
 - Inferiorly placed foramen magnum
- Gracility
- High activity levels
- Chins
- Reduced canines; no Cp3 honing complex
- Precision grip
 - Straight phalanges
- High encephalization quotient
- Globular cranium
- High capacity for plasticity (?)
- Long life histories with extended childhood & extended postreproductive period in females

Traits that 'make' us human



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The modern human body plan



Split from Neanderthals

2 Ma

1 Ma

0.5 Ma

The modern human body plan is established by 2 Ma

Turkana boy; 1.5 Ma
Homo erectus



Body plan established

2 Ma

Sima de los Huesos; 0.4 Ma
Homo heidelbergensis



Split from Neanderthals

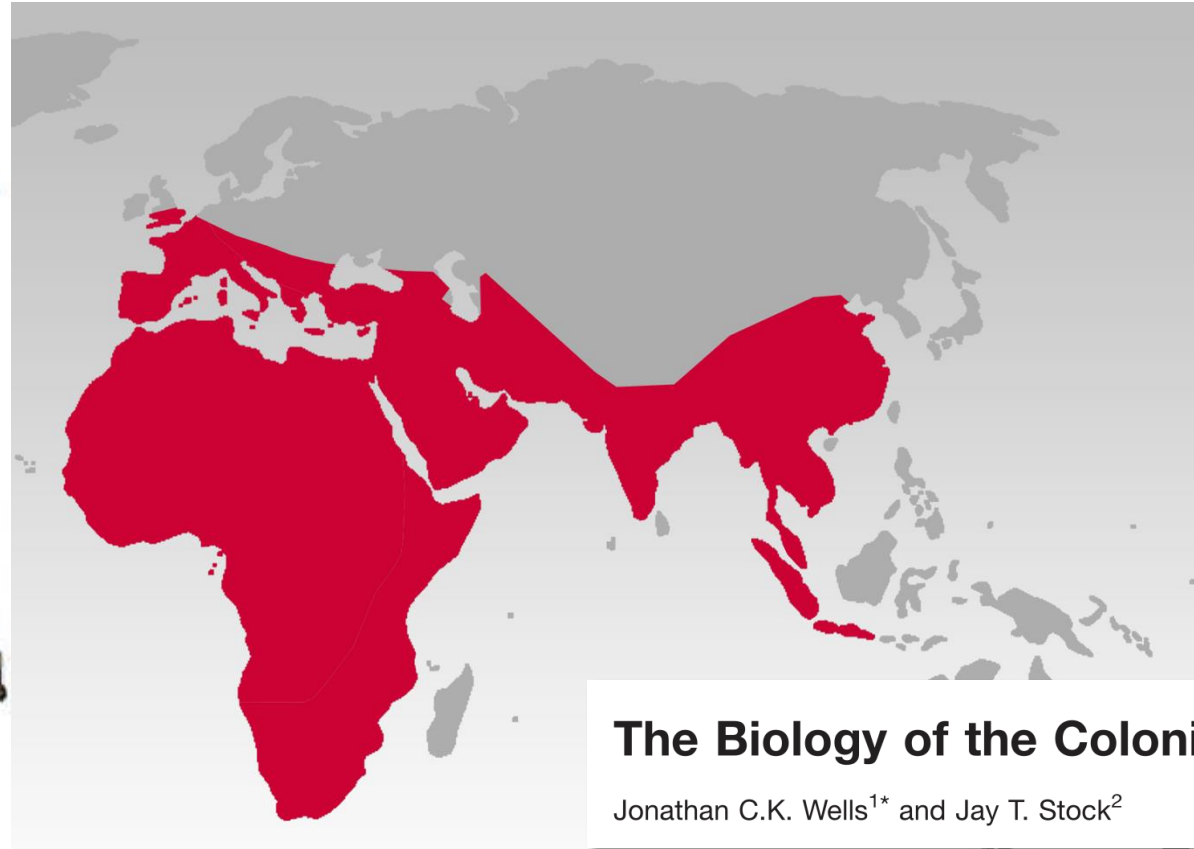
0.5 Ma

1 Ma

Did *Homo erectus* adapt to new environments through plasticity?



Body plan established



Split from Neanderthals

The modern human degree of plasticity *may* be established by 2 Ma



Various, including US immigrants and origin populations (relevant for controlling for genotype)



Various wild vs captive and wild populations OWM (relevant for controlling for genotype...somewhat)

How do known local conditions covary with morphology?

- In **humans** and **OWMs**, morphology is affected by environment
 - This pattern remains when controlling for sex & ancestry
- But... **non-human primates** show more robust responses to environmental change than humans do
 - Humans not so plastic after all?
 - Or an issue with samples?
 - Ancestry controlled for in humans
 - 'Pest' primates

Antón et al. (2016) *Phil Trans B*

Body plan established

Split from Neanderthals

The modern human degree of plasticity *may* be established by 2 Ma



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Various wild vs captive and wild populations OWM (relevant for controlling for genotype...somewhat)



Fossil hominins

- *Homo erectus*
- Neanderthals
- *Homo naledi*
- *Homo heidelbergensis*

Homo erectus:



Neanderthals:



Some matching attempted

- Dmanisi: more predators
- Zhoukoudian: fewer predators

Results:

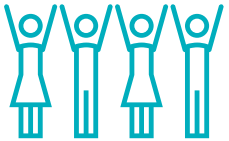
- *Homo erectus* shows more variation between its 'populations' than Neanderthals
- *Very brief* mention of lower variation than modern humans

Antón et al. (2016) *Phil Trans B*

Body plan established

Split from Neanderthals

The modern human degree of plasticity *may* be established by 2 Ma



Various, including US immigrants and origin populations (relevant for controlling for genotype)



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Fossil hominins

- *Homo erectus*
- Neanderthals
- *Homo naledi*
- *Homo heidelbergensis*

My view: so many confounds

- *Are humans more or less plastic than other primates?*
 - Poor sampling strategy: in humans *and* nonhumans, this sample is not controlled for genetics
- *Does Homo erectus respond plastically to its environment, thus buffering its genome from selection?*
 - No natural experiment available to control for genetic adaptation
 - Environmental context poorly known
 - Some populations sampled span a long time; others don't

Antón et al. (2016) *Phil Trans B*

Body plan established; maybe also developmentally plastic responses

Split from Neanderthals

2 Ma

1 Ma

0.5 Ma

Conclusions

- Most traits that are relevant for understanding modern human biology evolve far before the hominin lineage. Arguably, to really understand modern human biology, take a veterinary medicine class!
- The 'modern human package' - small face positioned beneath a globular cranium, reduced brow ridges, increased overall skeletal gracility - evolved gradually after the split with Neanderthals
 - The evolution of these traits likely are the consequence of a combination of processes that include adaptation to heat stress, self-domestication, and selection for sociality
- The modern human body plan appears in the fossil record around 2 Ma, and it may be the case that modern human levels of phenotypic plasticity existed at this time too.